Exercises and Solutions

Theory of Complex analysis

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Real-linear maps $\mathbb{R}^2 \to \mathbb{R}^2$ viewed as $\mathbb{C} \to \mathbb{C}$

Problem 0.1. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a real-linear map. Regard T as a map $\mathbb{C} \to \mathbb{C}$ via $(x, y) \leftrightarrow x+iy$. Show that there exist unique $A, B \in \mathbb{C}$ such that

$$T(z) = Az + B\overline{z} \qquad (\forall\, z \in \mathbb{C}),$$

and that T is complex differentiable (i.e. $\mathbb{C}$-linear) iff $B = 0$.

Background and definitions

- A map $T : \mathbb{C} \to \mathbb{C}$ is *real-linear* if $T(\alpha z + \beta w) = \alpha T(z) + \beta T(w)$ for all $z, w \in \mathbb{C}$ and all *real* α, β.
- It is *complex-linear* (holomorphic and linear) if the same holds for complex α, β, equivalently $T(iz) = iT(z)$ for all z.
- Every $z = x + iy$ has the conjugate $\bar{z} = x - iy$. Note that $x = \frac{z+\bar{z}}{2}$ and $y = \frac{z-\bar{z}}{2i}$.
- The *Wirtinger operators* are $\dfrac{\partial}{\partial z} = \dfrac{1}{2}\left(\dfrac{\partial}{\partial x} - i\dfrac{\partial}{\partial y}\right)$ and $\dfrac{\partial}{\partial \bar{z}} = \dfrac{1}{2}\left(\dfrac{\partial}{\partial x} + i\dfrac{\partial}{\partial y}\right)$. A C^1 map F is holomorphic iff $\partial F/\partial \bar{z} = 0$.

Solution

Existence. Let $\alpha := T(1)$ and $\beta := T(i)$, viewed as complex numbers. Seek $A, B \in \mathbb{C}$ with $T(z) = Az + B\bar{z}$. Then

$$\alpha = T(1) = A + B, \qquad \beta = T(i) = A\,i + B\,\bar{i} = i(A - B).$$

Solve

$$A + B = \alpha, \qquad A - B = -\,i\,\beta \quad \Longrightarrow \quad A = \frac{\alpha - i\beta}{2}, \qquad B = \frac{\alpha + i\beta}{2}.$$

Define A, B by these formulas. For any $z = x + iy$,

$$Az + B\bar{z} = A(x + iy) + B(x - iy) = (A + B)x + i(A - B)y = \alpha x + \beta y = T(x) + T(iy) = T(z),$$

since T is real-linear. Hence $T(z) = Az + B\bar{z}$ for all z.

Uniqueness. If $Az+B\bar{z} = A'z+B'\bar{z}$ for all z, then plugging $z = 1$ and $z = i$ gives $A+B = A'+B'$ and $i(A - B) = i(A' - B')$, hence $A = A'$ and $B = B'$.

Complex differentiability $\iff$ $B = 0$. If $B = 0$, then $T(z) = Az$ is complex-linear (multiplication by A), hence holomorphic. Conversely, if T is complex-linear, then $T(i) = iT(1)$, i.e. $\beta = i\alpha$. From the formulas above,

$$B = \frac{\alpha + i\beta}{2} = \frac{\alpha + i(i\alpha)}{2} = \frac{\alpha - \alpha}{2} = 0.$$

Equivalently, writing $T(z) = u(x,y) + iv(x,y)$, one computes

$$\frac{\partial T}{\partial \bar{z}} = B,$$

so the Cauchy–Riemann condition $\partial T/\partial \bar{z} = 0$ is exactly $B = 0$.

□

Concrete formulas in terms of a real matrix

If T has real matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ so that $T(x+iy) = (ax+by) + i(cx+dy)$, then

$$A = \frac{a+d}{2} + i\,\frac{c-b}{2}, \qquad B = \frac{a-d}{2} + i\,\frac{c+b}{2}.$$

Thus T is complex-linear iff $b = -c$ and $a = d$ (i.e. the matrix is $\left(\begin{smallmatrix} a & -t \\ t & a \end{smallmatrix}\right)$, multiplication by $a + it$).

Examples

1. **Identity:** $T(z) = z$. Here $A = 1$, $B = 0$ (holomorphic).
2. **Rotation-scaling:** $T(z) = \lambda z$ with $\lambda \in \mathbb{C}$. Then $A = \lambda$, $B = 0$ (holomorphic).
3. **Complex conjugation:** $T(z) = \bar{z}$. Here $A = 0$, $B = 1$ (not holomorphic).
4. **Real projection:** $T(z) = \mathrm{Re}(z) = \frac{z+\bar{z}}{2}$. Then $A = B = \frac{1}{2}$ (not holomorphic).
5. **General linear example:** $T(x+iy) = (2x+y) + i(3x-4y)$ has $A = \frac{2-4}{2} + i\frac{3-1}{2} = -1 + i$, $\quad B = \frac{2+4}{2} + i\frac{3+1}{2} = 3 + 2i$ (not holomorphic since $B \neq 0$).

Remarks

The decomposition $T(z) = Az + B\bar{z}$ is the *polarization* of real-linear maps. The coefficient $A = \partial T/\partial z$ is the complex (holomorphic) part and $B = \partial T/\partial \bar{z}$ is the antiholomorphic part. Holomorphic linear maps are exactly those with vanishing antiholomorphic part.

Exercises on $T(z) = Az + B\bar{z}$ (with solutions)

1. (**Recover A, B from a real matrix**) Let T have real matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ via $T(x+iy) = (ax+by) + i(cx+dy)$. Derive A, B in $T(z) = Az + B\bar{z}$.

   Solution. Compare coefficients using $z = x + iy,\ \bar{z} = x - iy$:

   $$A = \frac{a+d}{2} + i\,\frac{c-b}{2}, \qquad B = \frac{a-d}{2} + i\,\frac{c+b}{2}.$$

2. (**Quick diagnostics**) For each matrix, compute (A, B) and decide if T is complex-linear.

   $$\text{(a)}\ \left(\begin{smallmatrix} 2 & 1 \\ -1 & 2 \end{smallmatrix}\right) \quad \text{(b)}\ \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) \quad \text{(c)}\ \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right).$$

   Solution. (a) $A = 2 + i(-1-1)/2 = 2 - i,\ B = \frac{2-2}{2} + i\frac{-1+1}{2} = 0 \Rightarrow$ holomorphic. (b) $A = 1 + i\,0 = 1,\ B = 0 + i\,1 = i \Rightarrow$ not holomorphic. (c) $A = \frac{0+0}{2} + i\,\frac{1-1}{2} = 0,\ B = \frac{0-0}{2} + i\,\frac{1+1}{2} = i \Rightarrow$ not holomorphic.

3. (**Matrix from A, B**) Given $T(z) = Az + B\bar{z}$ with $A = \alpha + i\beta$, $B = \gamma + i\delta$ ($\alpha, \beta, \gamma, \delta \in \mathbb{R}$), show the real matrix is
$$\begin{pmatrix} \alpha+\gamma & -\beta+\delta \\ \beta+\delta & \alpha-\gamma \end{pmatrix}.$$
Solution. Expand $Az + B\bar{z}$ with $z = x + iy$ and collect (x, y) coefficients.

4. (**CR in matrix form**) Show T is complex-linear iff $a = d$ and $b = -c$. *Solution.* From Ex. 1, $B = 0 \iff a = d,\ c = -b$, i.e. matrix $\left(\begin{smallmatrix} a & -t \\ t & a \end{smallmatrix}\right)$.

5. (**Wirtinger derivatives**) For $T(z) = Az + B\bar{z}$, compute $\partial T/\partial z$ and $\partial T/\partial \bar{z}$. *Solution.* $\dfrac{\partial T}{\partial z} = A, \qquad \dfrac{\partial T}{\partial \bar{z}} = B.$

6. (**Composition law**) Let $T(z) = Az + B\bar{z}$ and $S(z) = Cz + D\bar{z}$. Show
$$S \circ T(z) = (CA + D\overline{B})\, z \ + \ (CB + D\overline{A})\, \bar{z}.$$
Deduce that the set of real-linear maps $\mathbb{C} \to \mathbb{C}$ is closed under composition, and $S \circ T$ is complex-linear iff $CB + D\overline{A} = 0$. *Solution.* Note $\overline{T(z)} = \overline{A}\bar{z} + \overline{B}z$ and compute $S(T(z)) = C\,T(z) + D\,\overline{T(z)}$; read off the $z, \bar{z}$ coefficients.

7. (**Jacobian, determinant, and orientation**) Show the Jacobian determinant of T equals
$$\det DT = |A|^2 - |B|^2.$$
Conclude that T is orientation-preserving iff $|A| > |B|$, reversing iff $|A| < |B|$, and singular iff $|A| = |B|$.

   Solution. Using Ex. 3's matrix and a short computation: $\det = (\alpha+\gamma)(\alpha-\gamma) - (-\beta+\delta)(\beta+\delta) = (\alpha^2+\beta^2) - (\gamma^2+\delta^2) = |A|^2 - |B|^2$.

8. (**Operator norm and singular values**) Show the maximal and minimal stretch factors (singular values) of T are
$$\sigma_{\max} = |A| + |B|, \qquad \sigma_{\min} = \big||A| - |B|\big|.$$
Solution (sketch). Write $T(z) = A\big(z + \frac{B}{A}\bar{z}\big)$ if $A \neq 0$; by rotating/scaling the domain and range one reduces to $A \geq 0$, $B \geq 0$ real. Then T acts by stretching along two orthogonal directions with factors $A \pm B$, giving the claim. General case follows by unitary conjugation.

Worked examples and illustrations

Example A (Rotation–scaling; holomorphic). Let $T(z) = \lambda z$ with $\lambda = 1.3e^{i\pi/6}$. Then $A = \lambda$, $B = 0$. The image of the unit circle is a circle of radius 1.3 rotated by $\pi/6$, and T preserves angles and orientation.

Example B (Conjugation). Let $T(z) = \bar{z}$. Then $A = 0$, $B = 1$. The unit circle is fixed as a set, but orientation is reversed. T is not complex-differentiable since $B \neq 0$.

Example C (Real shear). Let T have real matrix $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. By the formulas,
$$A = \frac{1+1}{2} + i\,\frac{0-1}{2} = 1 - \tfrac{i}{2}, \qquad B = \frac{1-1}{2} + i\,\frac{0+1}{2} = \tfrac{i}{2}.$$
The image of the unit circle is an ellipse tilted by the shear. Since $B \neq 0$, T is not holomorphic. The Jacobian is $|A|^2 - |B|^2 = (1^2 + (\frac{1}{2})^2) - (\frac{1}{2})^2 = 1 > 0$, so orientation is preserved.

Example D (General mixed map). Take $A = 1 + \frac{1}{2}i$, $B = 0.6 - 0.2i$. Then

$$\det DT = |A|^2 - |B|^2 = (1^2 + \tfrac{1}{4}) - (0.6^2 + 0.2^2) = 1.25 - 0.40 = 0.85 > 0,$$

so T is invertible and orientation-preserving, but not holomorphic ($B \neq 0$). Singular values are $\sigma_{\max} = |A| + |B| \approx \sqrt{1.25} + \sqrt{0.40}$ and $\sigma_{\min} = ||A| - |B||$, giving principal stretches of the ellipse.

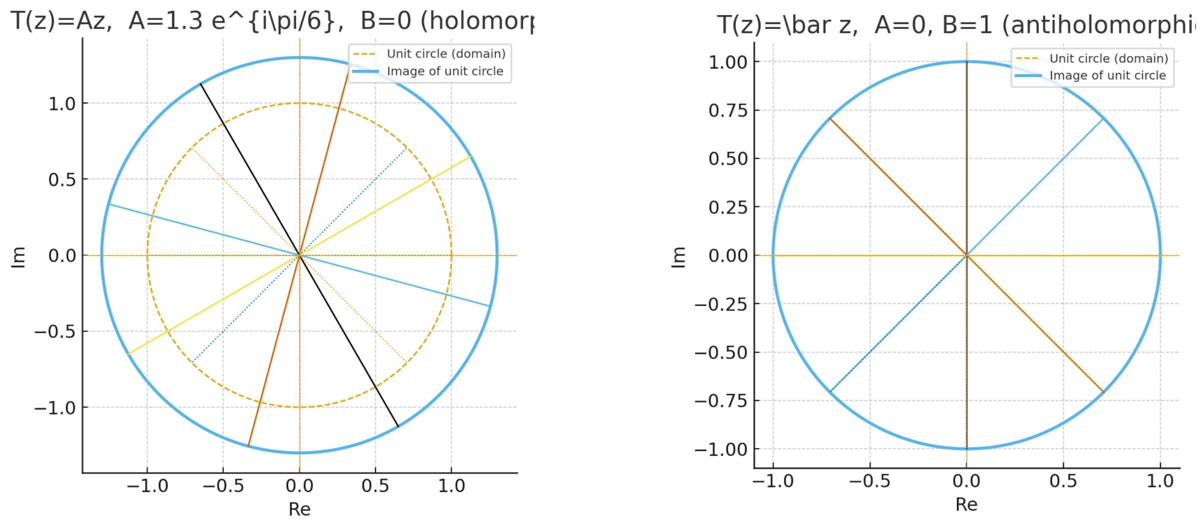

Figure 1: Left: $T(z) = 1.3e^{i\pi/6}z$ (holomorphic). Right: $T(z) = \bar{z}$ (orientation-reversing).

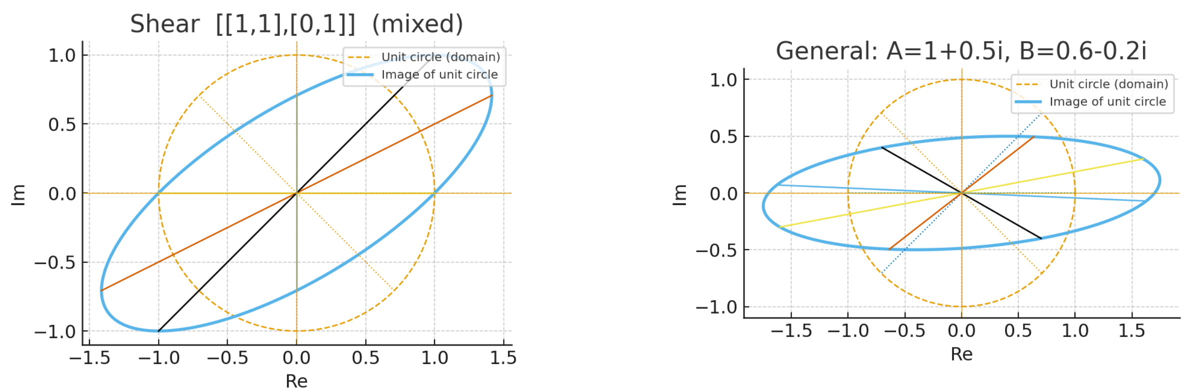

Figure 2: Left: real shear $[[1, 1], [0, 1]]$. Right: general mixed case $A = 1 + 0.5i$, $B = 0.6 - 0.2i$.

Notes. For any $T(z) = Az + B\bar{z}$, the image of the unit circle is an ellipse whenever $|A| \neq |B|$; it degenerates to a segment/point when $|A| = |B|$. Holomorphicity is equivalent to $B = 0$; antiholomorphicity (composition with conjugation) corresponds to $A = 0$.

Holomorphic functions with constrained parts

Problem 2. (i) Let $f : U \to \mathbb{C}$ be holomorphic on a domain U. Show that f is constant if any one of its real part, modulus, or argument is constant.

(ii) Find all entire functions of the form $f(x+iy) = u(x) + i\,v(y)$ with u, v real-valued.
(iii) Find all entire functions whose real part is $x^3 - 3xy^2$.

Solution. *(i) One part constant $\Rightarrow$ f constant.* Write $f = u + iv$. If $\Re f = u$ is constant, then $u_x = u_y = 0$; the CR equations $u_x = v_y$, $u_y = -v_x$ give $v_x = v_y = 0$, hence f is constant.

If $|f|$ is constant, either $f \equiv 0$ or $f(U)$ lies on a circle, which is not open; by the Open Mapping Theorem, f must be constant. (Alternatively: for holomorphic f, $f_x = f'$ and $f_y = if'$, so from $|f|^2$ constant $0 = \partial_x |f|^2 = 2\Re(f'\bar{f})$ and $0 = \partial_y |f|^2 = -2\Im(f'\bar{f})$, hence $f'\bar{f} = 0$, so $f' \equiv 0$ and f is constant.)

If $\arg f$ is constant, $f(U)$ lies on a ray, not open; the Open Mapping Theorem again forces f to be constant.

(ii) Form $u(x) + iv(y)$. Here $u_y = 0$ and $v_x = 0$. CR yields $u_x = v_y =: c$ (a real constant). Hence $u(x) = cx + a$, $v(y) = cy + b$ with $a, b \in \mathbb{R}$, $c \in \mathbb{R}$, and

$$f(z) = u(x) + iv(y) = (a + ib) + c(x + iy) = (a + ib) + c\,z.$$

Thus all such entire functions are $f(z) = \alpha + cz$ with $\alpha \in \mathbb{C}$ and $c \in \mathbb{R}$.

(iii) Prescribed real part $x^3 - 3xy^2$. Since $\Re(z^3) = x^3 - 3xy^2$, any entire f with this real part satisfies $f - z^3$ has zero real part and hence is a purely imaginary constant. Therefore

$$\boxed{f(z) = z^3 + i\beta, \quad \beta \in \mathbb{R}.}$$

A function satisfying CR at 0 but not differentiable there

Problem 3. Define $f : \mathbb{C} \to \mathbb{C}$ by $f(0) = 0$ and

$$f(z) = \frac{(1+i)x^3 - (1-i)y^3}{x^2 + y^2}, \qquad z = x + iy \neq 0.$$

Show that f satisfies the Cauchy–Riemann equations at 0 but is not differentiable there.

Solution. Write

$$(1+i)x^3 - (1-i)y^3 = (x^3 - y^3) + i(x^3 + y^3),$$

so for $(x, y) \neq (0, 0)$

$$u(x,y) = \frac{x^3 - y^3}{x^2 + y^2}, \qquad v(x,y) = \frac{x^3 + y^3}{x^2 + y^2},$$

and put $u(0,0) = v(0,0) = 0$.

CR at $(0,0)$. Compute partials at the origin via limits along the axes:

$$u_x(0,0) = \lim_{h\to 0} \frac{u(h,0) - 0}{h} = \lim_{h\to 0} \frac{h^3/h^2}{h} = 1, \quad u_y(0,0) = \lim_{k\to 0} \frac{u(0,k) - 0}{k} = \lim_{k\to 0} \frac{-k^3/k^2}{k} = -1,$$

$$v_x(0,0) = \lim_{h\to 0} \frac{v(h,0) - 0}{h} = \lim_{h\to 0} \frac{h^3/h^2}{h} = 1, \quad v_y(0,0) = \lim_{k\to 0} \frac{v(0,k) - 0}{k} = \lim_{k\to 0} \frac{k^3/k^2}{k} = 1.$$

Hence $u_x(0,0) = v_y(0,0) = 1$ and $u_y(0,0) = -v_x(0,0) = -1$, so the Cauchy–Riemann equations hold at 0.

Not differentiable at 0. If f were complex-differentiable at 0, then $f'(0) = u_x(0,0)+iv_x(0,0) = 1+i$ and $\lim_{z\to 0} \frac{f(z)}{z}$ would exist and equal $1+i$. However,

$$\frac{f(x)}{x} \to 1+i \quad (y=0), \qquad \frac{f(x+ix)}{x+ix} = \frac{ix}{x(1+i)} = \frac{1+i}{2} \quad (y=x),$$

so the limit depends on the path. Therefore f is not differentiable at 0. □

CR vs. complex differentiability (what is enough and what is not)

- **CR at a point: necessary, not sufficient.** If $f = u + iv$ is complex-differentiable at z_0, then the partials exist at z_0 and

  $$u_x(z_0) = v_y(z_0), \qquad u_y(z_0) = -\,v_x(z_0).$$

  These are *necessary*. However, they are *not sufficient* in general.

- **Why not sufficient?** Existence of the four partials at z_0 satisfying CR at z_0 does not guarantee real (Fréchet) differentiability as a map $\mathbb{R}^2 \to \mathbb{R}^2$. Hence the complex difference quotient may fail to have a limit. (Problem 3 gives a standard counterexample: CR hold at 0 but $f'(0)$ does not exist.)

- **Sufficient add-ons.** The CR equations *do* become sufficient under mild regularity:

  1. If f is C^1 in a neighborhood of z_0 and the CR equations hold there, then f is holomorphic in that neighborhood.
  2. If f is real-differentiable at z_0 and the CR equations hold at z_0, then f is complex-differentiable at z_0.

- **Wirtinger formulation.** If f is real-differentiable at z_0, then

  $$f \text{ is complex-differentiable at } z_0 \iff \frac{\partial f}{\partial \bar z}(z_0) = 0, \qquad \text{and then } f'(z_0) = \frac{\partial f}{\partial z}(z_0).$$

  If $f \in C^1$ on a neighborhood, the single condition $\partial f/\partial \bar z \equiv 0$ there implies holomorphicity.

Limits with the principal logarithm and complex powers

Problem 4. (i) Let Log denote the principal branch of log on $\mathbb{C} \setminus (-\infty, 0]$. For $z \in \mathbb{C}$, show that $n \operatorname{Log}(1 + z/n)$ is defined for n sufficiently large and that $n \operatorname{Log}(1 + z/n) \to z$. Deduce $\lim_{n\to\infty} \big(1 + \frac{z}{n}\big)^n = e^z$.

(ii) Defining $z^\alpha := \exp(\alpha \operatorname{Log} z)$ for $\alpha \in \mathbb{C}$ and $z \notin \mathbb{R}_{\le 0}$, show $\frac{d}{dz} z^\alpha = \alpha z^{\alpha-1}$. Does $(zw)^\alpha = z^\alpha w^\alpha$ always hold?

Solution. *(i) Limit of* $n \operatorname{Log}(1 + z/n)$. For fixed z, choose n large so that $1 + \frac{z}{n}$ lies in a small disc around 1 disjoint from $(-\infty, 0]$, hence Log is holomorphic there. Since

$$\lim_{u\to 0} \frac{\operatorname{Log}(1+u)}{u} = 1,$$

we get, with $u = z/n$,

$$n \operatorname{Log}\Big(1 + \frac{z}{n}\Big) = z\, \frac{\operatorname{Log}(1 + z/n)}{z/n} \ \longrightarrow\ z.$$

Therefore

$$\lim_{n\to\infty}\Big(1+\frac{z}{n}\Big)^n = \lim_{n\to\infty} \exp\!\Big(n\,\mathrm{Log}\big(1+\tfrac{z}{n}\big)\Big) = \exp(z) = e^z.$$

(ii) Derivative and multiplicative law. On $\mathbb{C}\setminus(-\infty,0]$, Log is holomorphic with $(\mathrm{Log}\,z)' = 1/z$, hence

$$\frac{d}{dz}z^\alpha = \frac{d}{dz}\exp(\alpha\,\mathrm{Log}\,z) = \exp(\alpha\,\mathrm{Log}\,z)\,\alpha\,\frac{1}{z} = \alpha z^{\alpha-1}.$$

For z, w in the domain of Log, in general

$$\mathrm{Log}(zw) = \mathrm{Log}\,z + \mathrm{Log}\,w - 2\pi i\,k, \qquad k\in\mathbb{Z},$$

so

$$(zw)^\alpha = \exp\!\big(\alpha\,\mathrm{Log}(zw)\big) = z^\alpha w^\alpha\, e^{-2\pi i\alpha k}.$$

Thus $(zw)^\alpha = z^\alpha w^\alpha$ holds when no branch jump occurs ($-\pi < \mathrm{Arg}\,z + \mathrm{Arg}\,w < \pi$), or when $\alpha\in\mathbb{Z}$; otherwise it can fail.

Counterexample. With $\alpha = \frac{1}{2}$ and $z = w = e^{3\pi i/4}$, $\mathrm{Log}\,z = \mathrm{Log}\,w = i\,3\pi/4$ but $\mathrm{Log}(zw) = -i\pi/2$, hence $z^\alpha w^\alpha = e^{i\,3\pi/4} \neq e^{-i\pi/4} = (zw)^\alpha$.

Principal logarithm and argument

Definition. For $z\in\mathbb{C}\setminus(-\infty,0]$ write $z = re^{i\theta}$ with $r = |z| > 0$ and $\theta = \mathrm{Arg}\,z \in (-\pi,\pi]$. The principal logarithm is

$$\mathrm{Log}\,z = \ln|z| + i\ \mathrm{Arg}\,z = \ln r + i\theta,$$

holomorphic on $\mathbb{C}\setminus(-\infty,0]$ with $(\mathrm{Log}\,z)' = 1/z$.

Basic properties.

- $\Re(\mathrm{Log}\,z) = \ln|z|$ and $\Im(\mathrm{Log}\,z) = \mathrm{Arg}\,z$.
- $\exp(\mathrm{Log}\,z) = z$ for all z off the branch cut.
- In general $\mathrm{Log}(zw) = \mathrm{Log}\,z + \mathrm{Log}\,w - 2\pi i\,k$ for some $k\in\mathbb{Z}$ chosen so that $\mathrm{Arg}(zw)\in(-\pi,\pi]$.

Geometry. Let $w = \mathrm{Log}\,z = u + iv$. Then:

- Circles $|z| = r$ map to vertical lines $u = \ln r$.
- Rays $\mathrm{Arg}\,z = \theta$ map to horizontal lines $v = \theta$.
- The branch cut $(-\infty,0]$ is where $v = \mathrm{Arg}\,z$ jumps from π to $-\pi$.

Geometry of $\mathrm{Log}\,z$ and z^α

A) Principal logarithm $w = \mathrm{Log}\,z = u + iv = \ln|z| + i\ \mathrm{Arg}\,z$**.**

- Ray $\mathrm{Arg}\,z = \theta_0$ maps to the horizontal line $v = \theta_0$.
- Circle $|z| = r_0$ maps to the vertical line $u = \ln r_0$.
- Annulus $r_1 < |z| < r_2$ maps to the vertical strip $\ln r_1 < u < \ln r_2$.
- Sector $\theta_1 < \mathrm{Arg}\,z < \theta_2$ maps to the horizontal strip $\theta_1 < v < \theta_2$.
- Multiplication $z\mapsto cz$ corresponds to the translation $(u,v)\mapsto(u+\ln|c|,\ v+\mathrm{Arg}\,c)$.

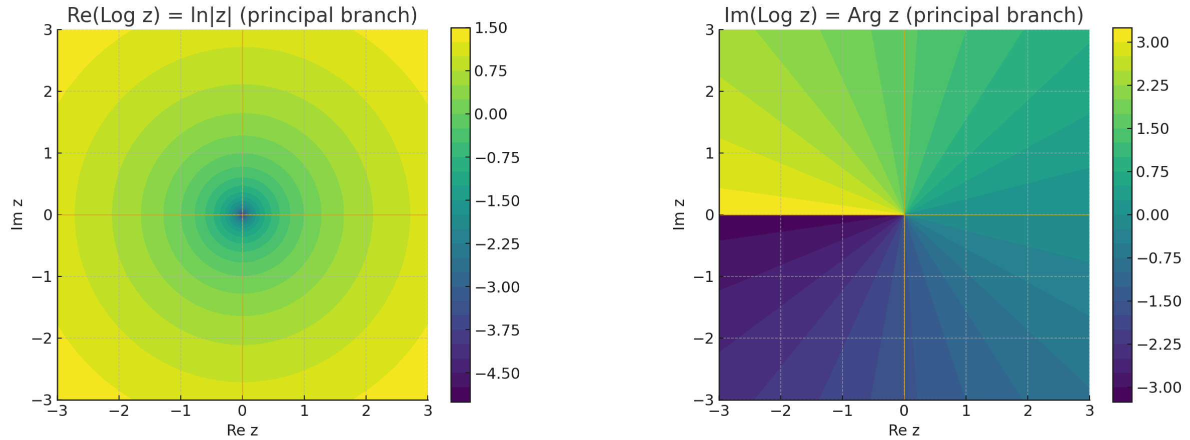

Figure 3: Left: $\Re(\operatorname{Log} z) = \ln|z|$. Right: $\Im(\operatorname{Log} z) = \operatorname{Arg} z$ on the principal branch.

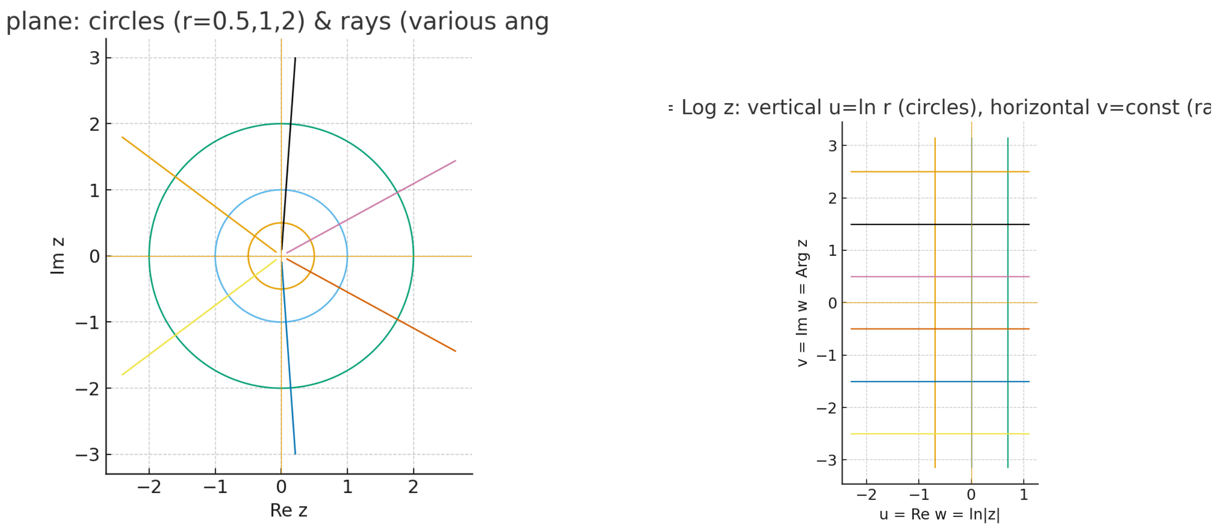

Figure 4: Circles and rays in the z-plane (left) straighten to vertical/horizontal lines under $w = \operatorname{Log} z$ (right).

B) Complex powers $w = z^\alpha$ with $\alpha = a + ib$. Writing $z = re^{i\theta}$ (principal $\theta = \operatorname{Arg} z$), we have

$$|w| = r^a e^{-b\theta}, \qquad \operatorname{Arg} w = a\theta + b \ln r.$$

Equivalently, $\operatorname{Log} w = \alpha \operatorname{Log} z$, i.e. the linear map $(u, v) \mapsto (au - bv,\ bu + av)$ in the (u, v)-plane followed by $w = e^{u+iv}$.

- Ray $\theta = \theta_0$ maps to a logarithmic spiral (to a ray if $b = 0$).
- Circle $r = r_0$ maps to a logarithmic spiral (to a circle of radius r_0^a if $b = 0$).
- Annulus maps to a region between two spirals (to a ring if $b = 0$).
- Sector maps to a spiral wedge (to a sector with angles scaled by a if $b = 0$).

Special cases.

- $\alpha = a \in \mathbb{R}$: rays $\mapsto$ rays (angle scaled by a); circles $\mapsto$ circles (radius $r \mapsto r^a$).
- $\alpha = ib$: rays $\mapsto$ families of circles; circles $\mapsto$ families of rays; generic curves $\mapsto$ spirals.

For $z \in \mathbb{C} \setminus (-\infty, 0]$,
$$\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z.$$

If $z = re^{i\theta}$ with $r = |z|$ and $\theta = \operatorname{Arg} z$, and $\alpha = a + ib \in \mathbb{C}$, then
$$z^\alpha = e^{\alpha \operatorname{Log} z} = r^a e^{-b\theta} \big[\cos\big(a\theta + b\ln r\big) + i \sin\big(a\theta + b\ln r\big)\big],$$
so
$$\Re(z^\alpha) = r^a e^{-b\theta} \cos\big(a\theta + b\ln r\big), \quad \Im(z^\alpha) = r^a e^{-b\theta} \sin\big(a\theta + b\ln r\big).$$

Geometry of $\operatorname{Log} z$ and z^α

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- Multiplication $z \mapsto cz$ corresponds to the translation $(u, v) \mapsto (u + \ln|c|,\ v + \operatorname{Arg} c)$.

B) Complex powers $w = z^\alpha$ with $\alpha = a + ib$. Writing $z = re^{i\theta}$ (principal $\theta = \operatorname{Arg} z$), we have
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- Sector maps to a spiral wedge (to a sector with angles scaled by a if $b = 0$).

Special cases.

- $\alpha = a \in \mathbb{R}$: rays $\mapsto$ rays (angle scaled by a); circles $\mapsto$ circles (radius $r \mapsto r^a$).
- $\alpha = ib$: rays $\mapsto$ families of circles; circles $\mapsto$ families of rays; generic curves $\mapsto$ spirals.

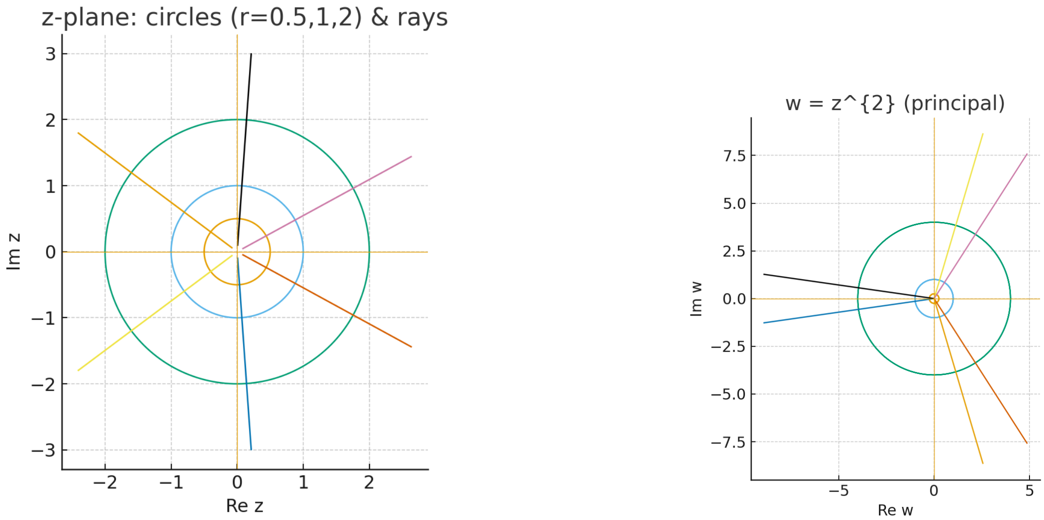

Figure 5: Left: circles and rays in the z-plane. Right: image under $w = z^2$ (principal branch).

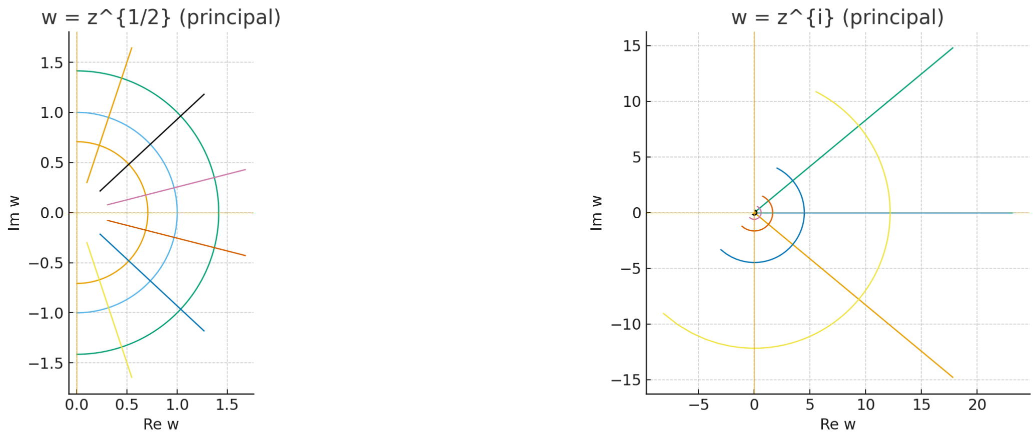

Figure 6: Left: $w = z^{1/2}$ (principal root). Right: $w = z^i$ (pure imaginary exponent).

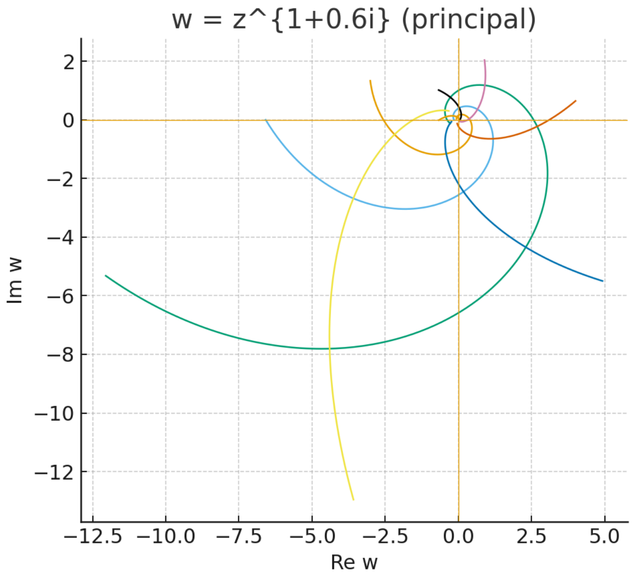

Figure 7: General complex exponent $w = z^{1+0.6i}$: rays and circles both map to logarithmic spirals.

Exponential, trig, and hyperbolic basics

Problem 5. (i) Verify directly that $e^z, \cos z, \sin z$ satisfy the Cauchy–Riemann equations everywhere.
(ii) Find all z with $|e^{iz}| > 1$, and describe the set where $|e^z| \le e^{|z|}$.
(iii) Find the zeros of $1 + e^z$ and of $\cosh z$.

Background. For $z = x + iy$,

$$e^z = e^x(\cos y + i \sin y), \qquad \sin z = \sin x \cosh y + i \cos x \sinh y, \qquad \cos z = \cos x \cosh y - i \sin x \sinh y.$$

If $f = u + iv$, the CR equations are $u_x = v_y$ and $u_y = -v_x$.

Solution. *(i) CR verification.*

$$\begin{array}{rll} e^z: & u = e^x \cos y,\ v = e^x \sin y & \Rightarrow u_x = e^x \cos y = v_y, \quad u_y = -e^x \sin y = -v_x. \\ \sin z: & u = \sin x \cosh y,\ v = \cos x \sinh y & \Rightarrow u_x = \cos x \cosh y = v_y, \quad u_y = \sin x \sinh y = -v_x. \\ \cos z: & u = \cos x \cosh y,\ v = -\sin x \sinh y & \Rightarrow u_x = -\sin x \cosh y = v_y, \quad u_y = \cos x \sinh y = -v_x. \end{array}$$

Hence each function satisfies CR everywhere and is entire.

(ii) Modulus questions. Let $z = x + iy$.

$$|e^{iz}| = |e^{ix-y}| = e^{-y} \quad \Rightarrow \quad |e^{iz}| > 1 \iff y < 0.$$

Also $|e^z| = e^x \le e^{\sqrt{x^2+y^2}} = e^{|z|}$ for all z, with equality iff $y = 0$ and $x \ge 0$.

(iii) Zeros.

$$1 + e^z = 0 \iff e^z = -1 \iff z = (2k+1)\pi i,\ k \in \mathbb{Z}.$$

$$\cosh z = \frac{e^z + e^{-z}}{2} = 0 \iff e^{2z} = -1 \iff 2z = (2k+1)\pi i \iff z = \frac{(2k+1)\pi i}{2},\ k \in \mathbb{Z}.$$

□

(ii) Sets defined by $|e^{iz}| > 1$ and $|e^z| \le e^{|z|}$. Write $z = x + iy$.

- $|e^{iz}| = |e^{i(x+iy)}| = |e^{ix-y}| = e^{-y}$. Thus

  $$|e^{iz}| > 1 \iff e^{-y} > 1 \iff y < 0,$$

  i.e. the **lower half-plane**. Equality $|e^{iz}| = 1$ occurs when $y = 0$.
- $|e^z| = |e^{x+iy}| = e^x$. Since $|z| = \sqrt{x^2 + y^2} \ge |x| \ge x$, we have

  $$|e^z| = e^x \ \le \ e^{|z|} \qquad \text{for all } z \in \mathbb{C}.$$

  Equality holds iff $|z| = x$, i.e. $y = 0$ and $x \ge 0$ (the **nonnegative real axis**).

Uniform convergence on complex subsets

Problem 6. Prove uniform convergence for:

$$\text{(a)} \quad \sum_{n=1}^{\infty} \sqrt{n}\, e^{-nz} \ \text{ on } \ \{z : 0 < r \le \Re z\}, \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{2^n}{z^n + z^{-n}} \ \text{ on } \ \{z : |z| \le r < \tfrac{1}{2}\}.$$

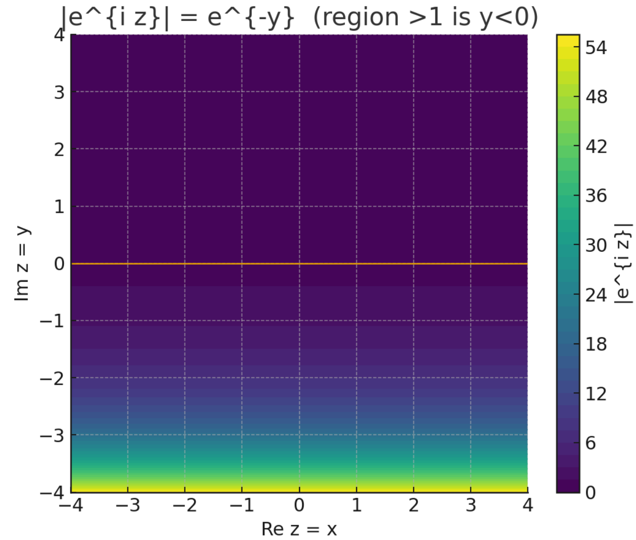

Figure 8: Heatmap of $|e^{iz}| = e^{-y}$. The region $|e^{iz}| > 1$ is $y < 0$; boundary $|e^{iz}| = 1$ is $y = 0$.

Tool (Weierstrass M-test). If $|f_n(z)| \le M_n$ on a set E and $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E.

Solution. *(a)* For $\Re z \ge r > 0$,

$$\left|\sqrt{n}\, e^{-nz}\right| = \sqrt{n}\, e^{-n\Re z} \le \sqrt{n}\, e^{-nr}.$$

Since $\sum_{n\ge 1} \sqrt{n}\, e^{-nr}$ converges (e.g. ratio or root test), the M-test yields uniform convergence on $\{z : \Re z \ge r\}$.

(b) Rewrite

$$\frac{2^n}{z^n + z^{-n}} = \frac{2^n z^n}{1 + z^{2n}},$$

which is entire (in particular, defined at $z = 0$). For $|z| \le r < \frac{1}{2}$,

$$\left|\frac{2^n z^n}{1 + z^{2n}}\right| \le \frac{2^n r^n}{1 - |z|^{2n}} \le \frac{(2r)^n}{1 - r^2}.$$

Because $2r < 1$, the series $\sum (2r)^n$ converges; hence, by the M-test, $\sum \dfrac{2^n}{z^n + z^{-n}}$ converges uniformly on $\{|z| \le r\}$. $\square$

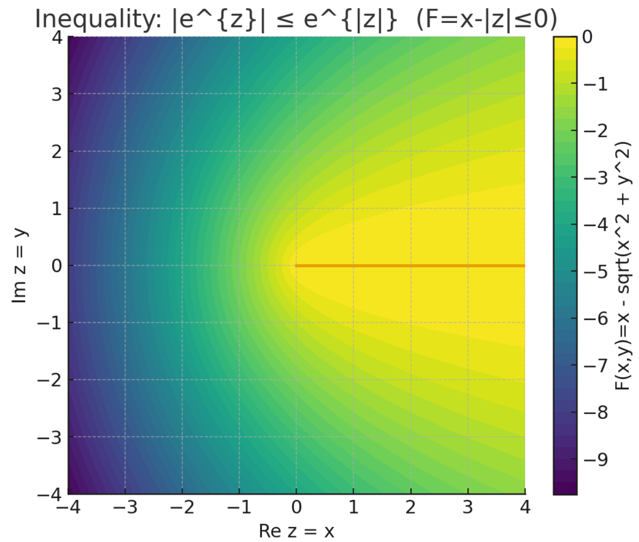

Figure 9: Heatmap of $F(x,y) = x - \sqrt{x^2+y^2} = \log|e^z| - |z|$. The inequality $|e^z| \leq e^{|z|}$ holds everywhere ($F \leq 0$); equality only on $y = 0$, $x \geq 0$.

Problem 7 (Conformal equivalences and a harmonic function). Find conformal equivalences between the following pairs of domains:

(i) The sector
$$S_1 = \{\, z \in \mathbb{C} : -\tfrac{\pi}{4} < \operatorname{Arg} z < \tfrac{\pi}{4} \,\}$$
and the open unit disc
$$\mathbb{D} = D(0,1) = \{\, w \in \mathbb{C} : |w| < 1 \,\}.$$

(ii) The lens
$$L = \{\, z \in \mathbb{C} : |z-1| < \sqrt{2} \text{ and } |z+1| < \sqrt{2} \,\}$$
and $\mathbb{D}$.

(iii) The strip
$$S = \{\, z \in \mathbb{C} : 0 < \Im z < 1 \,\}$$
and the quadrant
$$Q = \{\, w \in \mathbb{C} : \Re w > 0,\ \Im w > 0 \,\}.$$

By considering a suitable bounded solution of Laplace's equation $u_{xx} + u_{yy} = 0$ on S, find a non-constant harmonic function on Q which is constant on its boundary axes.

Conformal equivalences for classical domains

Background. Möbius maps send circles/lines to circles/lines and preserve angles. If $W_\alpha = \{z :\ |\operatorname{Arg} z| < \alpha\}$, then $z \mapsto z^{\pi/(2\alpha)}$ maps W_α onto a half–plane. Right half–plane $\{\Re w > 0\}$ maps to $\mathbb{D}$ via $C(w) = (w-1)/(w+1)$. The exponential $z \mapsto e^{\pi z}$ maps the strip $0 < \Im z < 1$ onto the upper half–plane.

(i) Sector to unit disk. Let $S_1 = \{-\frac{\pi}{4} < \operatorname{Arg} z < \frac{\pi}{4}\}$. Then $w = z^2$ gives $-\frac{\pi}{2} < \operatorname{Arg} w < \frac{\pi}{2} \iff \Re w > 0$. Composing with C,
$$F_1(z) = \frac{z^2 - 1}{z^2 + 1}$$
is a conformal bijection $S_1 \to \mathbb{D}$.

(ii) Lens to unit disk. Let $L = \{|z-1| < \sqrt{2}\} \cap \{|z+1| < \sqrt{2}\}$. Both boundary circles pass through i and $-i$ and meet orthogonally. Set $M(z) = \dfrac{z-i}{z+i}$, which sends $i \mapsto 0$ and $-i \mapsto \infty$, hence each boundary circle maps to a straight line through the origin, and $M(L)$ is a sector of opening $\pi/2$. Then $(\cdot)^2$ maps this sector to the right half–plane, and C maps to $\mathbb{D}$:
$$F_2(z) = \frac{\big(M(z)\big)^2 - 1}{\big(M(z)\big)^2 + 1} = \frac{\left(\dfrac{z-i}{z+i}\right)^2 - 1}{\left(\dfrac{z-i}{z+i}\right)^2 + 1}.$$
This is a conformal equivalence $L \to \mathbb{D}$.

(iii) Strip to quadrant. Let $S = \{0 < \Im z < 1\}$ and $Q = \{\Re w > 0,\ \Im w > 0\}$. Then
$$F_3(z) = e^{\frac{\pi}{2} z}$$
is a conformal bijection $S \to Q$, with inverse $F_3^{-1}(w) = \frac{2}{\pi} \operatorname{Log} w$ (principal branch).

Harmonic function on Q constant on the axes. Define $u(x+iy) = y$ on S; it is harmonic and bounded with $u = 0$ on $\Im z = 0$ and $u = 1$ on $\Im z = 1$. Push forward by F_3:

$$v(w) = u\big(F_3^{-1}(w)\big) = \Im\big(\tfrac{2}{\pi}\operatorname{Log} w\big) = \frac{2}{\pi}\operatorname{Arg} w, \qquad w \in Q.$$

Then v is harmonic on Q, satisfies $0 < v < 1$, and

$$v(x) = 0 \ \ (x > 0,\ \text{real axis}), \qquad v(iy) = 1 \ \ (y > 0,\ \text{imaginary axis}).$$

$\square$

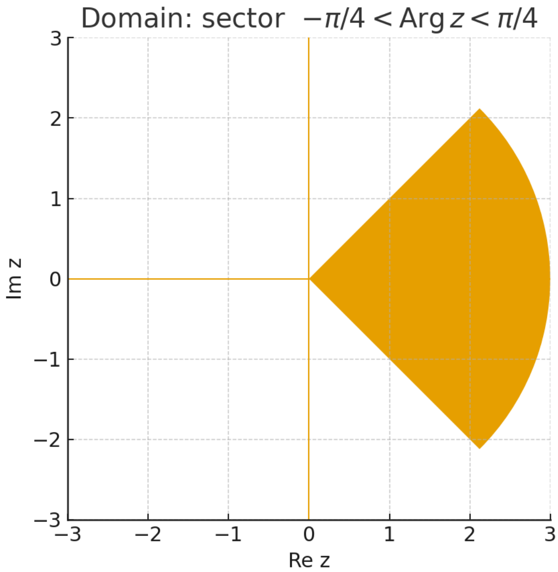

Figure 10: Sector domain $-\pi/4 < \operatorname{Arg} z < \pi/4$.

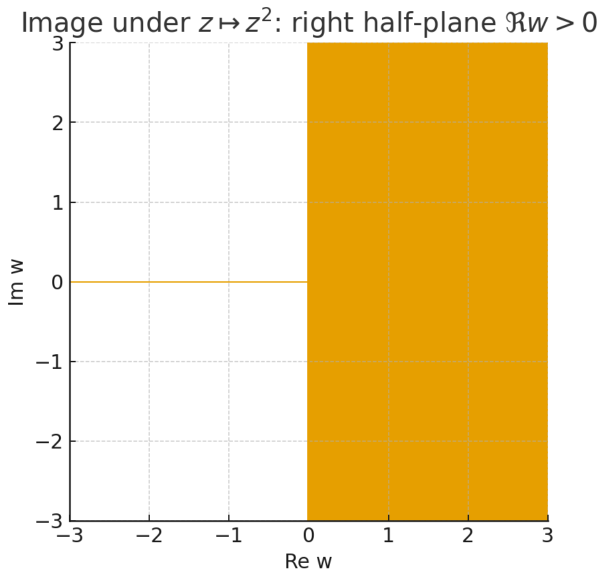

Figure 11: Image under $z \mapsto z^2$: right half–plane $\Re w > 0$.

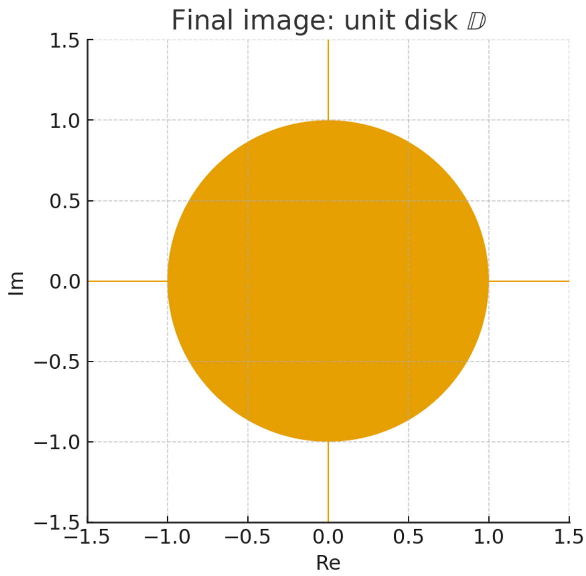

Figure 12: After Cayley transform: unit disk $\mathbb{D}$.

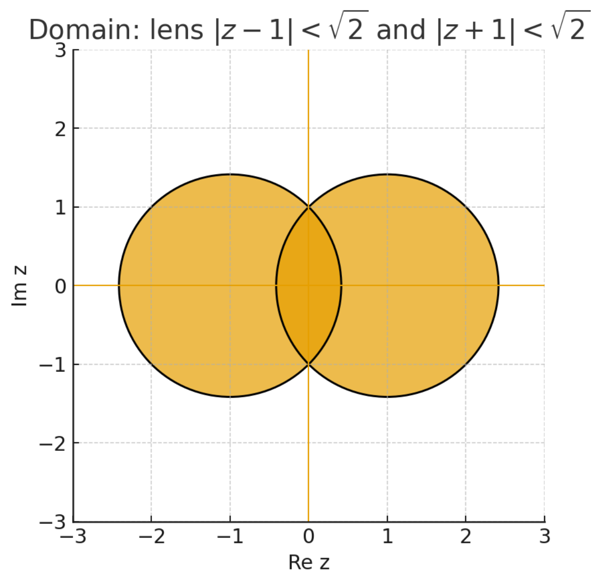

Figure 13: Lens domain $\{|z-1|<\sqrt{2}\}\cap\{|z+1|<\sqrt{2}\}$.

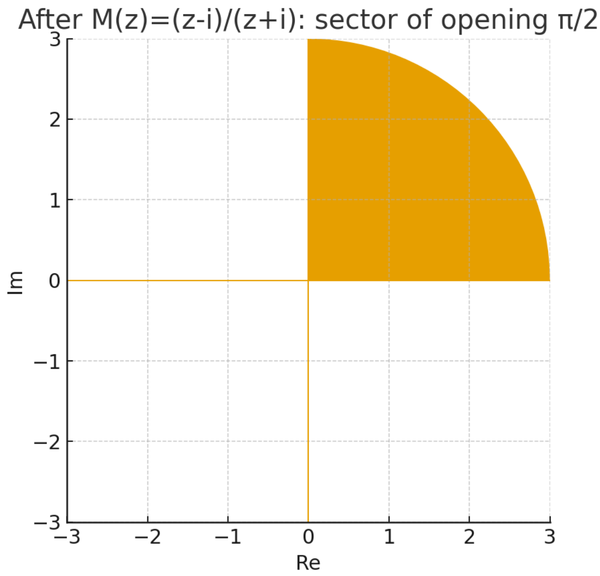

Figure 14: After $M(z) = (z - i)/(z + i)$: sector of opening $\pi/2$.

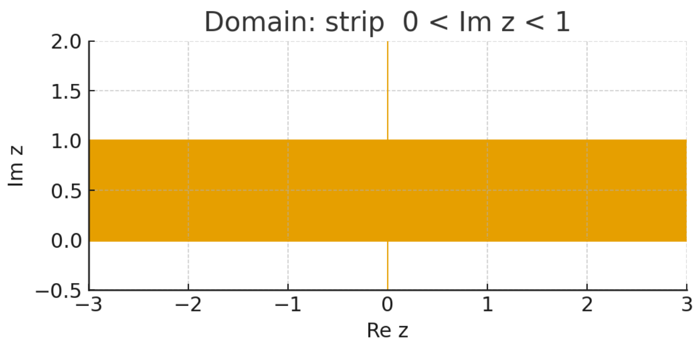

Figure 15: Strip domain $0 < \Im z < 1$.

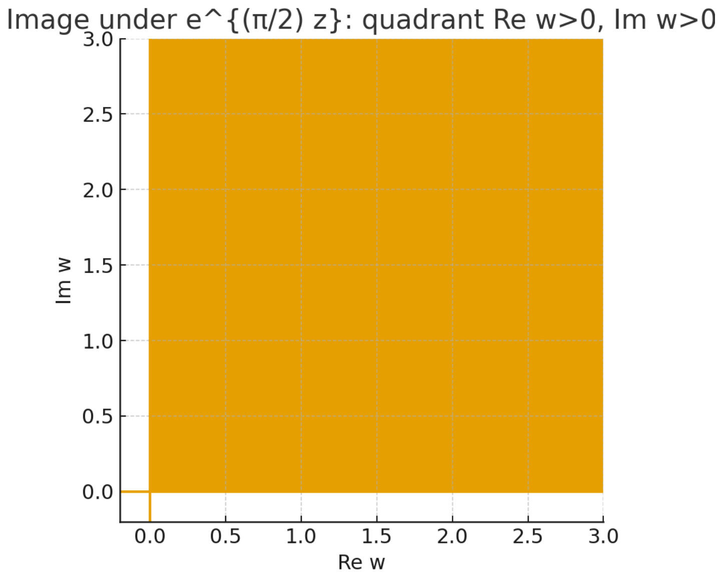

Figure 16: Image under $e^{(\pi/2)z}$: quadrant $\Re w > 0,\ \Im w > 0$.

Automorphisms of $\mathbb{D}$, annuli from circles, and strips to annuli

Problem 8.

(i) Show that a Möbius map sends the unit disk $\mathbb{D} = \{z : |z| < 1\}$ onto itself iff

$$\phi_{a,\lambda}(z) = \lambda \frac{z-a}{1-\bar{a}\,z}, \qquad |a| < 1,\ |\lambda| = 1.$$

(ii) Find a Möbius map taking the region between the circles $\{|z| = 1\}$ and $\{|z-1| = 5/2\}$ to an annulus $\{1 < |w| < R\}$.

(iii) Find a conformal map from an infinite strip onto an annulus. Can such a map be the restriction of a Möbius map?

Background. A Möbius map $M(z) = \dfrac{az+b}{cz+d}$ maps lines/circles to lines/circles and preserves angles. For $|a| < 1$ and $|z| < 1$,

$$|1-\bar{a}z|^2 - |z-a|^2 = (1-|a|^2)(1-|z|^2) > 0,$$

hence $\left|\dfrac{z-a}{1-\bar{a}z}\right| < 1$. If $g : \mathbb{D} \to \mathbb{D}$ is holomorphic with $g(0) = 0$, Schwarz lemma gives $g(z) = \lambda z$, $|\lambda| \le 1$. Given two disjoint circles, the map $F_{p,q}(z) = \dfrac{z-p}{z-q}$ with p, q the limiting points of their coaxal family sends them to concentric circles.

Solution. *(i)* The maps $\phi_{a,\lambda}$ send $\mathbb{D}$ to itself and are bijective with inverse $\phi_{-\lambda a,\ \bar{\lambda}}$. Conversely, let f be an automorphism of $\mathbb{D}$ and set $a = f^{-1}(0)$. Then $\psi(z) = \dfrac{z-a}{1-\bar{a}z}$ maps $a \mapsto 0$, so $g = f \circ \psi^{-1}$ is an automorphism with $g(0) = 0$. By Schwarz, $g(z) = \lambda z$ with $|\lambda| = 1$, hence $f(z) = \lambda \dfrac{z-a}{1-\bar{a}z}$.

(ii) The two circles are disjoint and nested. Their coaxal limiting points are the real numbers

$$p = \frac{2}{7}, \qquad q = -\frac{2}{3},$$

the two solutions of $|x| = \frac{1}{2.5}|x-1|$ on $\mathbb{R}$. Then

$$F(z) = \frac{z-p}{z-q}$$

maps $\{|z| = 1\}$ and $\{|z-1| = 5/2\}$ to concentric circles $|w| = \rho_1$ and $|w| = \rho_2$ (Apollonius property). Consequently the region between them maps to the annulus $\{\min(\rho_1, \rho_2) < |w| < \max(\rho_1, \rho_2)\}$. Rescale to unit inner radius by $G(z) = F(z)/\rho_{\text{inner}}$, obtaining

$$G: \quad \{1 < |w| < R\}, \qquad R = \rho_{\text{outer}}/\rho_{\text{inner}}.$$

(Any equivalent choice obtained by post/pre-composing with disk automorphisms is acceptable.)

(iii) Let $S_h = \{z : 0 < \Im z < h\}$. The map

$$W(z) = \exp\!\Big(\tfrac{2\pi}{h}\, z\Big)$$

is a conformal bijection $S_h \to \{1 < |w| < e^{2\pi}\}$, since $|W(z)| = \exp\!\big(\frac{2\pi}{h}\Im z\big) \in (1, e^{2\pi})$. This cannot be the restriction of a Möbius map: a Möbius map sends parallel boundary lines of a strip to two lines or to two circles meeting at one point, whereas the annulus is bounded by two *concentric* circles (which do not meet). $\square$

Problem 9. Calculate $\displaystyle\int_\gamma z \sin z \, dz$ where γ is the straight line joining 0 to i.

Solution. Since $z \sin z$ is entire, the integral is path independent. An antiderivative is

$$F(z) = \int z \sin z \, dz = -z \cos z + \sin z,$$

because $F'(z) = -\cos z + z \sin z + \cos z = z \sin z$. Hence

$$\int_\gamma z \sin z \, dz = F(i) - F(0) = \big(\sin i - i \cos i\big) - 0.$$

Using $\sin i = i \sinh 1$ and $\cos i = \cosh 1$,

$$\sin i - i \cos i = i(\sinh 1 - \cosh 1) = -\frac{i}{e}.$$

Therefore

$$\boxed{\int_\gamma z \sin z \, dz = -\frac{i}{e}.}$$

Problem 10. Show that the following functions do not have antiderivatives on the indicated domains:

$$\text{(a) } f(z) = \frac{1}{z} - \frac{1}{z-1} \quad \text{on } \{\, 0 < |z| < 1 \,\}, \qquad \text{(b) } g(z) = \frac{z}{1+z^2} \quad \text{on } \{\, 1 < |z| < \infty \,\}.$$

Problem 10. Show that the following functions do not have antiderivatives on the indicated domains:

$$\text{(a) } f(z) = \frac{1}{z} - \frac{1}{z-1} \quad \text{on } \{\, 0 < |z| < 1 \,\}, \qquad \text{(b) } g(z) = \frac{z}{1+z^2} \quad \text{on } \{\, 1 < |z| < \infty \,\}.$$

Background (tools). We collect the standard criteria we will use.

- **Path-independence / primitives.** If h is holomorphic on a domain U and has a primitive H on U (i.e. $H' = h$), then

$$\int_\gamma h(z) \, dz = 0 \quad \text{for every closed curve } \gamma \subset U,$$

  and conversely, if every closed integral vanishes on U, then h has a primitive on U.

- **Residue theorem (basic form).** Let h be meromorphic on an open set containing a positively oriented simple closed curve γ and its interior, with no poles on γ. Then

$$\int_\gamma h(z) \, dz = 2\pi i \sum_{a \in \mathrm{Int}(\gamma)} \mathrm{Res}(h; a).$$

  For a *simple pole* at a, $\mathrm{Res}(h; a) = \lim_{z \to a} (z-a) h(z)$. (Equivalently, if $h = p/q$ with p, q holomorphic and $q'(a) \neq 0$, then $\mathrm{Res}(h; a) = \dfrac{p(a)}{q'(a)}$.)

- **No global log $\Rightarrow$ no primitive.** A holomorphic h on U has a holomorphic logarithm $\log h$ on U iff $\int_\gamma h'(z)/h(z) \, dz = 0$ for every closed $\gamma \subset U$ (equivalently, h has no zeros with nonzero winding number around loops in U). In particular, if a holomorphic expression forces a nontrivial jump of argument around a hole, then a single-valued primitive involving log cannot exist on U.

- **Winding number version.** If γ is a positively oriented circle $|z-z_0|=r$, then
$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z-z_0}=1.$$
More generally, for any closed γ and $a\notin\gamma$, $\dfrac{1}{2\pi i}\displaystyle\int_\gamma \frac{dz}{z-a}$ equals the winding number of γ around a.

Solution. *(a)* $f(z)=\dfrac{1}{z}-\dfrac{1}{z-1}$ *on* $0<|z|<1$. Let γ_r be the positively oriented circle $|z|=r$ with $0<r<1$. Then $\gamma_r\subset\{0<|z|<1\}$, and the only singularity of f in the interior is $z=0$ (simple pole with residue 1). Thus, by the residue theorem,
$$\int_{\gamma_r} f(z)\,dz=2\pi i\ \mathrm{Res}(f;0)=2\pi i\neq 0.$$
If f had a primitive on the annulus $0<|z|<1$, every closed integral would vanish—a contradiction. Hence no primitive exists there.

(b) $g(z)=\dfrac{z}{1+z^2}$ *on* $1<|z|<\infty$. Let Γ_R be the circle $|z|=R>1$. Then $\Gamma_R\subset\{1<|z|<\infty\}$. The function g has simple poles at $\pm i$ (both lie inside Γ_R and off the curve). Using the simple-pole formula with $p(z)=z$, $q(z)=1+z^2$,
$$\mathrm{Res}(g;i)=\frac{p(i)}{q'(i)}=\frac{i}{2i}=\frac{1}{2},\qquad \mathrm{Res}(g;-i)=\frac{p(-i)}{q'(-i)}=\frac{-i}{-2i}=\frac{1}{2}.$$
Hence
$$\int_{\Gamma_R} g(z)\,dz=2\pi i\Big(\tfrac{1}{2}+\tfrac{1}{2}\Big)=2\pi i\neq 0,$$
so g cannot have a primitive on the exterior domain $\{1<|z|<\infty\}$.

Alternative viewpoint for (b). If a primitive G existed, we could write $G'=\dfrac{z}{1+z^2}$, whence locally $G=\frac{1}{2}\log(1+z^2)+\text{const}$. But on the punctured exterior domain, any loop around the unit circle winds once around the *zeros* $\pm i$ of $1+z^2$, so the argument of $1+z^2$ changes by 2π and $\log(1+z^2)$ changes by $2\pi i$—contradicting single-valuedness. Thus no global primitive exists. $\square$

Solution. *(a)* Let γ_r be the circle $|z|=r$ with $0<r<1$. Then $\gamma_r\subset\{0<|z|<1\}$ and the only singularity of f inside γ_r is $z=0$ with residue 1. Hence
$$\int_{\gamma_r} f(z)\,dz=2\pi i\ \mathrm{Res}(f;0)=2\pi i\neq 0.$$
Therefore f has no antiderivative on $\{0<|z|<1\}$.

(b) Let Γ_R be the circle $|z|=R>1$. Then $\Gamma_R\subset\{1<|z|<\infty\}$ and g is meromorphic on and inside Γ_R with simple poles at $\pm i$ (not on the curve). We have
$$\mathrm{Res}(g;i)=\frac{i}{2i}=\frac{1}{2},\qquad \mathrm{Res}(g;-i)=\frac{-i}{-2i}=\frac{1}{2},$$
so
$$\int_{\Gamma_R} g(z)\,dz=2\pi i\Big(\tfrac{1}{2}+\tfrac{1}{2}\Big)=2\pi i\neq 0.$$
Thus g has no antiderivative on $\{1<|z|<\infty\}$. $\square$

Lemma 1 (Path–independence and primitives)**.** *Let $U \subset \mathbb{C}$ be a domain and $h \in \mathcal{O}(U)$. The following are equivalent:*

1. *h has a primitive H on U;*
2. *$\int_\gamma h\,dz = 0$ for every closed, piecewise C^1 curve $\gamma \subset U$.*

In this case, $\int_\gamma h\,dz = H(\gamma(1)) - H(\gamma(0))$ for any path γ.

Lemma 2 (Residue theorem; simple pole formula)**.** *Let h be meromorphic on a neighborhood of a positively oriented simple closed curve γ with no poles on γ. Then*

$$\int_\gamma h(z)\,dz \;=\; 2\pi i \sum_{a\in \mathrm{Int}(\gamma)} \mathrm{Res}(h;a).$$

If $h = p/q$ with p, q holomorphic and $q(a) = 0$, $q'(a) \neq 0$ (simple pole), then $\mathrm{Res}(h;a) = \dfrac{p(a)}{q'(a)}$.

Lemma 3 (No global log $\Rightarrow$ no primitive)**.** *Let $U \subset \mathbb{C}$ be a domain and F holomorphic, nonvanishing on U. A holomorphic branch $\operatorname{Log} F$ exists on U iff $\int_\gamma \dfrac{F'(z)}{F(z)}\,dz = 0$ for every closed curve $\gamma \subset U$. In particular, if a loop in U winds nontrivially around a zero/pole of F from outside U, then no single-valued $\operatorname{Log} F$ exists on U.*

Corollary 1. *If $h(z) = \frac{1}{z-a}$ on $U = \mathbb{C} \setminus \{a\}$, then for the circle $\gamma(t) = a + re^{it}$, $0 \le t \le 2\pi$, we have $\int_\gamma h\,dz = 2\pi i$. Hence h has no primitive on U.*

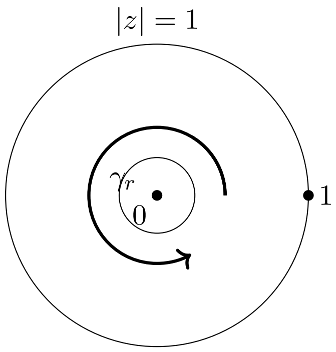

Figure 17: Part (a): $\gamma_r \subset \{0 < |z| < 1\}$ encloses 0 but not 1.

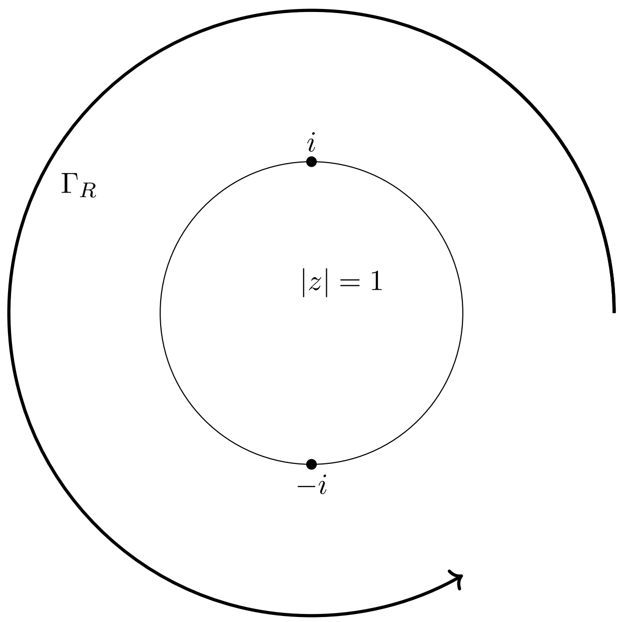

Figure 18: Part (b): $\Gamma_R \subset \{1 < |z| < \infty\}$ encloses the poles $\pm i$.

Problem 11. Let $U \subset \mathbb{C}$ be a domain and $u : U \to \mathbb{R}$ be C^2 and harmonic. Show that if $z_0 \in U$, then for any disk $D = D(z_0, r) \Subset U$ there exists a holomorphic $f : D \to \mathbb{C}$ with $\Re f = u$ on D. Show by an example that this need not hold globally.

Background. A C^2 real function u is *harmonic* if

$$\Delta u \;=\; u_{xx} + u_{yy} \;=\; 0.$$

Set the 1-form

$$\alpha \;=\; -\,u_y\,dx \;+\; u_x\,dy.$$

A short calculation gives

$$d\alpha \;=\; (u_{xx} + u_{yy})\,dx \wedge dy \;=\; \Delta u\,dx \wedge dy.$$

Hence if u is harmonic, α is *closed* ($d\alpha = 0$). On a simply connected domain (e.g. a disk), every closed 1-form is *exact*, so there exists v with $dv = \alpha$, i.e.

$$v_x = -u_y, \qquad v_y = u_x.$$

Then the Cauchy–Riemann equations hold for (u, v) and $f := u + iv$ is holomorphic with $\Re f = u$.

On non–simply connected domains a closed form need not be exact; a certificate is a nonzero loop integral

$$\oint_\gamma \alpha \;\neq\; 0 \qquad \text{for some closed curve } \gamma \subset U.$$

Solution. *Local existence on a disk.* Fix $z_0 \in U$ and $D = D(z_0, r) \Subset U$. Since u is harmonic, $d\alpha = 0$ on D. Because D is simply connected, there exists $v : D \to \mathbb{R}$ with $dv = \alpha$, i.e.

$$v_x = -u_y, \qquad v_y = u_x.$$

Hence the Cauchy–Riemann equations hold and

$$f := u + iv \quad \text{is holomorphic on } D \text{ with } \Re f = u.$$

Failure globally: a counterexample. Let $U = \mathbb{C} \setminus \{0\}$ and $u(z) = \log|z|$. Then $u \in C^\infty(U)$ and $\Delta u = 0$. If there were a holomorphic F on U with $\Re F = u$, then its imaginary part $v = \Im F$ would satisfy $dv = \alpha = -u_y\,dx + u_x\,dy$.

On the unit circle $\gamma(t) = e^{it}$, $0 \le t \le 2\pi$, one has

$$u_x = \frac{x}{x^2 + y^2}, \qquad u_y = \frac{y}{x^2 + y^2},$$

so along $|z| = 1$,

$$\oint_\gamma \alpha = \oint_\gamma (-y\,dx + x\,dy) = \int_0^{2\pi} \big(-\sin t \cdot (-\sin t) + \cos t \cdot \cos t \big)\,dt = \int_0^{2\pi} 1\,dt = 2\pi \;\neq\; 0.$$

Thus α is not exact on U, so no global v exists and u is *not* the real part of any holomorphic function on U. $\square$

Problem 1.

(i) Use Cauchy's integral formula to compute

$$\int_{|z|=1} \frac{e^{\alpha z}}{2z^2-5z+2}\,dz, \qquad \alpha \in \mathbb{C}.$$

(ii) By considering the real part of a suitable complex integral, show that for $r \in (0,1)$,

$$\int_0^{\pi} \frac{\cos(n\theta)}{1-2r\cos\theta+r^2}\,d\theta = \frac{\pi r^n}{1-r^2}, \qquad \int_0^{2\pi} \cos(\cos\theta)\cosh(\sin\theta)\,d\theta = 2\pi.$$

Background. Cauchy's integral formula: if f is holomorphic on and inside a simple closed curve γ, then $\displaystyle\int_\gamma \frac{f(z)}{z-a}\,dz = 2\pi i\, f(a)$ for a inside γ. Also, with $z = e^{i\theta}$ one has $d\theta = \frac{dz}{iz}$ and $1-2r\cos\theta+r^2 = (1-rz)(1-rz^{-1}) = \frac{(z-r)(1-rz)}{z}$.

Solution. *(i)* Factor $2z^2-5z+2 = 2(z-2)\big(z-\frac{1}{2}\big)$. On $|z|=1$ only $z=\frac{1}{2}$ lies inside. Hence

$$\int_{|z|=1} \frac{e^{\alpha z}}{2z^2-5z+2}\,dz = 2\pi i\cdot \operatorname{Res}\left(\frac{e^{\alpha z}}{2(z-2)\big(z-\frac{1}{2}\big)};\ \frac{1}{2}\right) = 2\pi i\cdot \frac{e^{\alpha/2}}{2(\frac{1}{2}-2)} = -\frac{2\pi i}{3}\,e^{\alpha/2}.$$

(ii-a) Let

$$I_n = \int_0^{2\pi} \frac{e^{in\theta}}{1-2r\cos\theta+r^2}\,d\theta.$$

With $z=e^{i\theta}$ and $d\theta = \frac{dz}{iz}$,

$$I_n = \oint_{|z|=1} \frac{z^n}{(z-r)(1-rz)}\,\frac{dz}{i}.$$

Only $z=r$ lies inside, giving $\operatorname{Res} = \dfrac{r^n}{1-r^2}$. Thus

$$I_n = 2\pi\,\frac{r^n}{1-r^2}, \qquad \int_0^{2\pi} \frac{\cos(n\theta)}{1-2r\cos\theta+r^2}\,d\theta = \Re I_n = \frac{2\pi r^n}{1-r^2}.$$

Halving the interval,

$$\int_0^{\pi} \frac{\cos(n\theta)}{1-2r\cos\theta+r^2}\,d\theta = \frac{\pi r^n}{1-r^2}.$$

(ii-b) Using $\cos(\cos\theta)\cosh(\sin\theta) = \frac{1}{2}\big(e^{ie^{i\theta}}+e^{-ie^{i\theta}}\big)$,

$$\int_0^{2\pi} \cos(\cos\theta)\cosh(\sin\theta)\,d\theta = \frac{1}{2}\oint_{|z|=1} \frac{e^{iz}+e^{-iz}}{iz}\,dz = \frac{1}{2i}\cdot 2\pi i\left(e^0+e^0\right) = 2\pi.$$

$\square$

Residues: definitions, equivalences, and tools. *Laurent series and the residue.* Let f be meromorphic with an isolated singularity at a. Then there is $\rho > 0$ and integers $m \geq 0$ and coefficients $\{c_k\}_{k=-m}^{\infty}$ such that

$$f(z) = \sum_{k=-m}^{\infty} c_k(z-a)^k, \qquad 0 < |z-a| < \rho.$$

Definition. The *residue* of f at a is the coefficient of $(z-a)^{-1}$:

$$\boxed{\operatorname{Res}(f;a) = c_{-1}}.$$

Equivalent characterizations. For any $0 < \varepsilon < \rho$,

$$\operatorname{Res}(f;a) = \frac{1}{2\pi i} \oint_{|z-a|=\varepsilon} f(z)\,dz.$$

If a is a simple pole (i.e. $m = 1$), then

$$\operatorname{Res}(f;a) = \lim_{z\to a} (z-a)\,f(z).$$

More generally, if a is a pole of order m,

$$\operatorname{Res}(f;a) = \frac{1}{(m-1)!} \lim_{z\to a} \frac{d^{m-1}}{dz^{m-1}}\big((z-a)^m f(z)\big).$$

Proof sketch of c_{-1}-rule. Write the Laurent series and integrate termwise on $|z-a| = \varepsilon$:

$$\oint (z-a)^k\,dz = \begin{cases} 0, & k \neq -1, \\ 2\pi i, & k = -1. \end{cases}$$

Thus $\displaystyle\oint f = 2\pi i\, c_{-1}$ and $\operatorname{Res}(f;a) = c_{-1}$.

Residue Theorem (workhorse). If f is meromorphic on and inside a positively oriented simple closed curve γ with no poles on γ, then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{a_k\in \operatorname{Int}(\gamma)} \operatorname{Res}(f;a_k).$$

Quick computation rules.

- **Simple pole formula.** If $f(z) = \dfrac{p(z)}{q(z)}$ with p, q holomorphic, $q(a) = 0$, $q'(a) \neq 0$, then

  $$\operatorname{Res}(f;a) = \frac{p(a)}{q'(a)}.$$

- **Analytic factor.** If g is holomorphic near a, then $\operatorname{Res}(g\,f;a) = g(a)\,\operatorname{Res}(f;a)$.
- **Pole at infinity.** If f is meromorphic on the Riemann sphere $\widehat{\mathbb{C}}$,

  $$\operatorname{Res}(f;\infty) := -\sum_{a\in\mathbb{C}} \operatorname{Res}(f;a), \qquad \oint_{|z|=R} f(z)\,dz \xrightarrow{R\to\infty} -2\pi i\,\operatorname{Res}(f;\infty),$$

  when the outer integral stabilizes.

Micro–examples (via c_{-1}).

- $f(z) = \dfrac{1}{(z-a)(z-b)}$ near a: expand $\dfrac{1}{z-b} = \dfrac{1}{a-b} + O(z-a)$, so

$$f(z) = \frac{1}{z-a} \cdot \frac{1}{a-b} + O(1) \;\Rightarrow\; \operatorname{Res}(f;a) = \frac{1}{a-b}.$$

- $f(z) = \dfrac{e^{\alpha z}}{(z-a)(z-b)}$ near a: $e^{\alpha z} = e^{\alpha a} + O(z-a)$, hence

$$f(z) = \frac{1}{z-a} \cdot \frac{e^{\alpha a}}{a-b} + O(1) \;\Rightarrow\; \operatorname{Res}(f;a) = \frac{e^{\alpha a}}{a-b}.$$

Unit-circle substitution for trig integrals. With $z = e^{i\theta}$, $d\theta = \dfrac{dz}{iz}$ and

$$1 - 2r\cos\theta + r^2 = (1-rz)(1-rz^{-1}) = \frac{(z-r)(1-rz)}{z}.$$

Only $z = r$ lies inside $|z| = 1$ if $0 < r < 1$, giving

$$\int_0^{2\pi} \frac{e^{in\theta}}{1 - 2r\cos\theta + r^2}\, d\theta = \oint_{|z|=1} \frac{z^n}{(z-r)(1-rz)}\, \frac{dz}{i} = 2\pi\, \frac{r^n}{1-r^2}.$$

Taking real parts yields $\displaystyle\int_0^{\pi} \frac{\cos(n\theta)}{1 - 2r\cos\theta + r^2}\, d\theta = \frac{\pi r^n}{1-r^2}$.

Problem 2. Find the Laurent expansion (in powers of z, i.e. about 0) of

$$\frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)}$$

in the regions $\{|z| < 1\}$, $\{1 < |z| < 2\}$, and $\{|z| \geq 2\}$.

Background. Partial fractions:

$$\frac{1}{(z-1)(z-2)} = -\frac{1}{z-1} + \frac{1}{z-2}.$$

Geometric expansions:

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n \quad (|w| < 1), \qquad \frac{1}{z-a} = \frac{1}{z} \cdot \frac{1}{1 - \frac{a}{z}} = \sum_{n=0}^{\infty} \frac{a^n}{z^{n+1}} \quad (|z| > |a|).$$

Solutions. *(A) For $|z| < 1$.*

$$-\frac{1}{z-1} = \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \qquad \frac{1}{z-2} = -\frac{1}{2}\frac{1}{1 - \frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}.$$

Hence

$$\boxed{\frac{1}{z^2 - 3z + 2} = \sum_{n=0}^{\infty} \Big(1 - 2^{-(n+1)}\Big) z^n, \quad |z| < 1.}$$

(B) For $1 < |z| < 2$.

$$-\frac{1}{z-1} = -\frac{1}{z}\frac{1}{1-\frac{1}{z}} = -\sum_{n=1}^{\infty} z^{-n}, \qquad \frac{1}{z-2} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}.$$

Thus

$$\boxed{\frac{1}{z^2-3z+2} = -\sum_{n=1}^{\infty} z^{-n} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \quad 1 < |z| < 2.}$$

(C) For $|z| > 2$.

$$-\frac{1}{z-1} = -\sum_{m=1}^{\infty} z^{-m}, \qquad \frac{1}{z-2} = \frac{1}{z}\frac{1}{1-\frac{2}{z}} = \sum_{m=1}^{\infty} \frac{2^{m-1}}{z^m}.$$

Therefore

$$\boxed{\frac{1}{z^2-3z+2} = \sum_{m=1}^{\infty} \left(2^{m-1}-1\right) z^{-m}, \quad |z| > 2.}$$

□

Problem 3. Classify the singularities of

$$\frac{z}{\sin z}, \qquad \sin\Big(\frac{\pi}{z^2}\Big), \qquad \frac{1}{z^2} + \frac{1}{z^2+1}, \qquad \frac{1}{z^2}\cos\Big(\frac{\pi z}{z+1}\Big).$$

Background. At an isolated singularity a, if the Laurent expansion $f(z) = \sum_{k=-m}^{\infty} c_k (z-a)^k$ has (i) $m = 0 \Rightarrow$ removable; (ii) finitely many negative terms with highest $(z-a)^{-m} \Rightarrow$ pole of order m; (iii) infinitely many negative terms $\Rightarrow$ essential.

(1) $\boldsymbol{z/\sin z}$. Near 0, $\sin z = z - \frac{z^3}{6} + O(z^5)$, so

$$\frac{z}{\sin z} = 1 + \frac{z^2}{6} + O(z^4),$$

hence $z = 0$ is **removable** (define value 1). At $z = n\pi$, $n \in \mathbb{Z} \setminus \{0\}$, $\sin z$ has simple zeros and the numerator is nonzero, so each $n\pi$ is a **simple pole**.

(2) $\boldsymbol{\sin(\pi/z^2)}$. Using $\sin w = \sum_{k\geq 0} \frac{(-1)^k w^{2k+1}}{(2k+1)!}$ with $w = \pi/z^2$,

$$\sin\Big(\frac{\pi}{z^2}\Big) = \sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k+1}}{(2k+1)!} z^{-4k-2},$$

which contains infinitely many negative powers. Thus $z = 0$ is an **essential singularity**. There are no other singularities.

(3) $\boldsymbol{1/z^2 + 1/(z^2+1)}$. At $z = 0$ one has a **pole of order** 2. At $z = \pm i$ (simple zeros of z^2+1) there are **simple poles**. No other singularities occur.

(4) $\boldsymbol{z^{-2}\cos\big(\pi z/(z+1)\big)}$. Let $h(z) = \dfrac{\pi z}{z+1}$. Near $z = 0$, $h(z) = \pi(z - z^2 + z^3 - \cdots)$, hence

$$\cos h(z) = 1 - \tfrac{1}{2}h(z)^2 + O(z^3) = 1 - \tfrac{\pi^2}{2}z^2 + O(z^3),$$

and

$$\frac{1}{z^2}\cos h(z) = \frac{1}{z^2} - \frac{\pi^2}{2} + O(z),$$

so $z = 0$ is a **pole of order** 2. At $z = -1$, h has a simple pole; since $\cos w = \frac{1}{2}(e^{iw} + e^{-iw})$, composition with a pole yields an **essential singularity** at $z = -1$. There are no other singularities. □

Problem 4. Let $f : \mathbb{C} \to \mathbb{C}$ be entire. Prove that f is constant if any one of the following holds:

(i) $\dfrac{f(z)}{z} \to 0$ as $|z| \to \infty$.

(ii) There exist $b \in \mathbb{C}$ and $\varepsilon > 0$ such that $|f(z) - b| > \varepsilon$ for all $z \in \mathbb{C}$.

(iii) Writing $f = u + iv$ with real u, v, one has $|u(z)| > |v(z)|$ for all $z \in \mathbb{C}$.

Background. We use: (a) *Liouville's theorem*: a bounded entire function is constant; (b) *Cauchy estimates*: if f is holomorphic on $|z| \le R$, then $|f^{(n)}(0)| \le \dfrac{n!}{R^n} \max_{|z|=R} |f(z)|$; (c) the Cayley map $w \mapsto \dfrac{w-1}{w+1}$ sends $\{\Re w > 0\}$ to the unit disk; (d) if $FG \equiv 0$ with F, G holomorphic on a connected set, then $F \equiv 0$ or $G \equiv 0$.

Solution. *(i) Sublinear growth.* Given $\eta > 0$, choose R_η so that $|f(z)| \le \eta|z|$ for $|z| \ge R_\eta$. For $R \ge R_\eta$,

$$\max_{|z|=R} |f(z)| \le \eta R.$$

Cauchy estimates at 0 yield

$$|f^{(n)}(0)| \le \frac{n!\,\eta R}{R^n} = \frac{n!\,\eta}{R^{n-1}} \quad (n \ge 1).$$

Letting $R \to \infty$ gives $f^{(n)}(0) = 0$ for $n \ge 2$, and $|f'(0)| \le \eta$ for all $\eta > 0$, hence $f'(0) = 0$. Thus f is constant.

(ii) Image avoids a disk. If $|f(z) - b| > \varepsilon$ for all z, then $g(z) = \dfrac{1}{f(z) - b}$ is entire and bounded by $1/\varepsilon$. By Liouville, g is constant, hence f is constant.

(iii) Real part dominates. If $f = u + iv$ with $|u| > |v|$ everywhere, then $\Re(f^2) = u^2 - v^2 > 0$, so $g := f^2$ maps $\mathbb{C}$ into the right half–plane. The Cayley transform $h = (g-1)/(g+1)$ is entire and satisfies $|h| < 1$, hence h is constant by Liouville, so $g \equiv c$ with $\Re c > 0$. Therefore $(f - \sqrt{c})(f + \sqrt{c}) \equiv 0$, and by analyticity one factor vanishes identically. Thus $f \equiv \sqrt{c}$ or $f \equiv -\sqrt{c}$, i.e. f is constant. □

Problem 6.

(i) Let f be entire. Show that f is a polynomial of degree $\le k$ iff there is $M > 0$ with

$$|f(z)| \le M(1 + |z|)^k \qquad (\forall z \in \mathbb{C}).$$

(ii) Show that an entire f is a polynomial of positive degree iff $|f(z)| \to \infty$ as $|z| \to \infty$.

(iii) Let f be analytic on $\mathbb{C}$ apart from finitely many poles. If there exists k such that $|f(z)| \le |z|^k$ for all sufficiently large $|z|$, prove that f is a rational function.

Background. *Cauchy estimates.* If f is holomorphic on $|z| \le R$, then

$$|f^{(n)}(0)| \le \frac{n!}{R^n} \max_{|z|=R} |f(z)|.$$

Poles at infinity. For $g(w) = f(1/w)$ near $w = 0$, a pole of order m at 0 means f has a pole at ∞ of order m. An entire function with a pole at ∞ is a polynomial (degree m). A function meromorphic on the Riemann sphere (only poles, finitely many) is rational.

Solution. *(i)* $\Leftarrow$*.* Assume $|f(z)| \le M(1+|z|)^k$. For $R > 0$, $\max_{|z|=R} |f(z)| \le M(1+R)^k$. Hence

$$|f^{(n)}(0)| \le \frac{n!}{R^n} M(1+R)^k \qquad (n \ge 0).$$

Fix $n > k$ and let $R \to \infty$: $(1+R)^k/R^n \to 0$, so $f^{(n)}(0) = 0$. Therefore the Taylor series truncates at degree $\le k$ and f is a polynomial.

(i) $\Rightarrow$*.* If $f(z) = \sum_{j=0}^k a_j z^j$, then $|f(z)| \le \sum_{j=0}^k |a_j||z|^j \le C(1+|z|)^k$.

(ii) $\Rightarrow$*.* If f is a nonconstant polynomial of degree $m \ge 1$, then $|f(z)| \sim |a_m||z|^m \to \infty$.

(ii) $\Leftarrow$*.* If $|f(z)| \to \infty$ as $|z| \to \infty$, then $g(w) = f(1/w)$ has a pole at $w = 0$, so f has a pole at ∞ and is therefore a polynomial of positive degree.

(iii). Let $g(w) = f(1/w)$. For small $|w|$,

$$|g(w)| = |f(1/w)| \le |1/w|^k = |w|^{-k},$$

so g has at most a pole of order $\le k$ at 0. Hence f has at most a pole at ∞. Together with the finitely many poles in $\mathbb{C}$, f is meromorphic on the Riemann sphere with only poles; thus f is rational. $\square$

Problem 7.

(i) (**Schwarz's Lemma**) Let f be analytic on $D(0,1)$ with $|f(z)| \le 1$ and $f(0) = 0$. Apply the maximum principle to $f(z)/z$ to show $|f(z)| \le |z|$. Show also that if $|f(w)| = |w|$ for some $w \ne 0$, then $f(z) = c\,z$ for some constant c.

(ii) Use Schwarz's Lemma to prove that any conformal equivalence $\Phi : D(0,1) \to D(0,1)$ is given by a Möbius transformation.

Background. If a holomorphic g on $0 < |z| < 1$ is bounded near 0, then g extends holomorphically to $z = 0$ (removable singularity). A non-constant holomorphic function does not attain a maximum of its modulus in the interior (maximum modulus principle). For $a \in D$, the map $\phi_a(z) = \dfrac{z-a}{1-\bar{a}\,z}$ is a biholomorphism $D \to D$ with $\phi_a(a) = 0$.

Solution. *(i)* Define

$$g(z) = \begin{cases} \dfrac{f(z)}{z}, & z \ne 0, \\ f'(0), & z = 0. \end{cases}$$

Because $|f(z)| \le 1$ and $f(0) = 0$, g is bounded near 0, hence extends holomorphically to D. For $0 < r < 1$, on $|z| = r$ we have $|g(z)| \le 1/r$. By the maximum modulus principle on each smaller disk, letting $r \uparrow 1$ gives $|g(z)| \le 1$ for all $z \in D$. Thus $|f(z)| \le |z|$ and $|f'(0)| = |g(0)| \le 1$.

If $|f(w)| = |w|$ for some $w \neq 0$, then $|g(w)| = 1$. By the maximum modulus principle g has constant modulus 1 on D, hence $g \equiv c$ with $|c| = 1$ and $f(z) = c\,z$.

(ii) Let $\Phi : D \to D$ be a conformal bijection and set $a = \Phi(0)$. Consider $F = \phi_a \circ \Phi$, where $\phi_a(z) = \dfrac{z-a}{1-\overline{a}\,z}$. Then $F : D \to D$ is holomorphic with $F(0) = 0$. By Schwarz's Lemma, $F(z) = \lambda z$ with $|\lambda| \leq 1$; since Φ is bijective, $|\lambda| = 1$. Therefore

$$\Phi(z) = \phi_a^{-1}(\lambda z) = \lambda\,\frac{z-a}{1-\overline{a}\,z},$$

a Möbius automorphism of the disk. $\square$

Conformal maps and conformal equivalence. *Conformal at a point.* A map $f : U \to \mathbb{C}$ is conformal at $z_0 \in U$ if it preserves angles between smooth curves through z_0 (with orientation). In complex analysis this is equivalent to

$$f \text{ holomorphic at } z_0 \quad \text{and} \quad f'(z_0) \neq 0.$$

Indeed, the first-order expansion $f(z) = f(z_0) + f'(z_0)(z-z_0) + o(|z-z_0|)$ shows f is locally "rotation + dilation". Writing $f = u + iv$, the Cauchy–Riemann equations at z_0 give

$$Df(z_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = |f'(z_0)| \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

so $Df(z_0)$ is a positive scalar times a rotation, hence angle-preserving and orientation-preserving.

Angle preservation formula. If $v_1, v_2 \in \mathbb{C}$ are tangent directions at z_0, then under f

$$\cos\angle(f_* v_1, f_* v_2) = \frac{\Re\big((f'(z_0)v_1)\overline{(f'(z_0)v_2)}\big)}{|f'(z_0)v_1|\,|f'(z_0)v_2|} = \frac{\Re(v_1\overline{v_2})}{|v_1||v_2|},$$

so the angle is unchanged.

Conformal equivalence. Domains $U, V \subset \mathbb{C}$ are *conformally equivalent* if there is a biholomorphism $f : U \to V$ (holomorphic and bijective with holomorphic inverse). Such an f is conformal on U, and f^{-1} is conformal on V.

Examples.

- Cayley transform $\phi(z) = \dfrac{z-i}{z+i}$: biholomorphism $\{\Im z > 0\} \leftrightarrow \mathbb{D}$.
- Exponential $w = e^{\pi z}$: biholomorphism $\{0 < \Im z < 1\} \leftrightarrow \{1 < |w| < e^{\pi}\}$.

Non-examples.

- $f(z) = z^n$ is not conformal at $z = 0$ for $n \geq 2$ since $f'(0) = 0$ (angles multiply by n).
- Conjugation $f(z) = \overline{z}$ reverses orientation (anti-conformal).

Problem 8 (restated).

(i) Let f be entire and satisfy $f(1/n) = 1/n$ for all $n \in \mathbb{N}$. Prove that $f(z) \equiv z$.

(ii) Let f be entire and satisfy $f(n) = n^2$ for all $n \in \mathbb{Z}$. Must f equal z^2? Give a justification or a counterexample.

(iii) Let f be holomorphic on $D(0,2)$. Show that there exists $n \in \mathbb{N}$ with $f(1/n) \neq 1/(n+1)$.

Background and definitions for Problem 8. *Holomorphic / entire.* A function f is holomorphic on a domain $U \subset \mathbb{C}$ if it is complex differentiable at every point of U. It is *entire* if it is holomorphic on all of $\mathbb{C}$. A *domain* is a nonempty, open, connected subset of $\mathbb{C}$.

Accumulation point. A sequence $(z_n) \subset U$ has an accumulation point $z_* \in U$ if $z_n \to z_*$ with $z_n \neq z_*$ eventually.

Isolated zeros and the identity theorem. If f is holomorphic on U and not identically zero, then its zeros have no accumulation point in U. If f, g are holomorphic on U and $f = g$ on a set with a limit point in U (e.g. $z_n \to z_* \in U$), then $f \equiv g$ on U. Equivalently, if a holomorphic f vanishes on such a set, then $f \equiv 0$.

An entire function vanishing on $\mathbb{Z}$. $\sin(\pi z)$ is entire and $\sin(\pi n) = 0$ for all $n \in \mathbb{Z}$. Thus for prescribed integer values, one may add $\sin(\pi z) \cdot H(z)$ (with H entire) without changing those values.

Removable singularities (Riemann). If f is holomorphic on $0 < |z - a| < r$ and bounded near a, then f extends holomorphically to $z = a$.

How these tools are used in Problem 8.

- **(i)** $g(z) = f(z) - z$ entire with $g(1/n) = 0$ and $1/n \to 0 \in \mathbb{C} \Rightarrow g \equiv 0$.
- **(ii)** Integers have no accumulation in $\mathbb{C}$; non-uniqueness via $\sin(\pi z)$.
- **(iii)** Zeros at $1/n \to 0$ in $D(0, 2) \setminus \{-1\}$ force identity with a meromorphic function that has a pole at -1 — contradiction.

Solution to Problem 8.

(i) Define
$$g(z) := f(z) - z.$$
Then g is entire and $g(1/n) = 0$ for all n. Since $1/n \to 0$ (an interior point), the identity theorem gives $g \equiv 0$, hence $f(z) \equiv z$.

(ii) Not necessarily. For example,
$$f(z) = z^2 + \sin(\pi z)$$
is entire and satisfies $f(n) = n^2$ for all $n \in \mathbb{Z}$, yet $f \not\equiv z^2$.

(iii) Suppose $f(1/n) = 1/(n + 1)$ for all n. Then
$$h(z) := f(z) - \frac{z}{z + 1}$$
is holomorphic on $D(0, 2) \setminus \{-1\}$, vanishes at $z = 1/n$ for all n, and the set $\{1/n\}$ accumulates at 0 in that domain. By the identity theorem, $h \equiv 0$ there, i.e. $f(z) = \dfrac{z}{z + 1}$ on $D(0, 2) \setminus \{-1\}$. But $\dfrac{z}{z + 1}$ has a pole at $z = -1$, whereas f is holomorphic there — a contradiction. Hence some n satisfies $f(1/n) \neq 1/(n + 1)$.

Problem 9 (Casorati–Weierstrass; restated). If f is holomorphic on $D(a, R) \setminus \{a\}$ with an essential singularity at a, then for any $b \in \mathbb{C}$ there exists a sequence $z_n \to a$ such that $f(z_n) \to b$.

Solution to Problem 9. If f omitted b near a, then $g = 1/(f-b)$ would be bounded near a, hence removable there by Riemann's theorem. Thus $f = b + 1/g$ would have at worst a pole/removable singularity at a, contradicting essentiality. *Example.* For $f(z) = e^{1/z}$, $a = 0$, $b = 2$, take

$$z_n = \frac{1}{\operatorname{Log} 2 + 2\pi i n} \quad \Rightarrow \quad e^{1/z_n} = 2.$$

Problem 10 (restated). Let $D \subset \mathbb{C}$ be simply connected and $0 \notin D$. Show that there exists a holomorphic function F on D with $F'(z) = 1/z$ and $e^{F(z)} = z$ for $z \in D$ (i.e. F is a branch of the logarithm on D).

Solution to Problem 10. Fix $z_0 \in D$ and define

$$F(z) = \int_{z_0}^{z} \frac{d\zeta}{\zeta}.$$

Since $1/\zeta$ is holomorphic on D and D is simply connected, the integral is path-independent; F is holomorphic with $F'(z) = 1/z$. Let $G(z) = e^{F(z)}/z$. Then $G'(z) = 0$, so G is constant. Choosing $F(z_0)$ so that $G \equiv 1$ yields $e^{F(z)} = z$ on D.

Problem 11 (restated). Consider the lacunary series

$$f(z) = \sum_{n=1}^{\infty} z^{2^n}.$$

Show that it defines an analytic function on $D(0,1)$ but admits no analytic continuation across any point of $|z| = 1$.

Solution to Problem 11. The series converges absolutely and uniformly on compact subsets of $D(0,1)$, so f is analytic there. Moreover, for $m \geq 1$,

$$(1 - z^{2^m}) f(z) = \sum_{j=1}^{m-1} z^{2^j},$$

so each 2^m-th root of unity is a singularity of f (unless trivially canceled). As the union $\bigcup_{m \geq 1} \{\zeta : \zeta^{2^m} = 1\}$ is dense on $|z| = 1$, $|z| = 1$ is a *natural boundary*: no analytic continuation is possible across any boundary point. □

Background for Problems 9–11. *Domains and basic classes.* A *domain* is a nonempty, open, connected subset of $\mathbb{C}$. A function is *holomorphic* on U if it is complex differentiable at every point of U. It is *entire* if holomorphic on $\mathbb{C}$. It is *meromorphic* on U if holomorphic on U except at isolated poles.

Isolated singularities. Let f be holomorphic on $0 < |z - a| < r$. Then a is an *isolated singularity* and exactly one holds:

- **Removable:** f is bounded near $a \Rightarrow f$ extends holomorphically to a (Riemann).
- **Pole:** $|f(z)| \to \infty$ as $z \to a$; equivalently, $(z-a)^m f(z)$ extends holomorphically with nonzero value at a for some $m \in \mathbb{N}$.

- **Essential:** neither removable nor pole. The Laurent expansion at a then has infinitely many negative powers.

Casorati–Weierstrass & Picard (context for Problem 9). If a is an essential singularity of f, then the image of any punctured neighborhood of a is dense in $\mathbb{C}$ (Casorati–Weierstrass). A stronger statement (Great Picard) says that near an essential singularity f attains every complex value, with at most one exception, infinitely often.

Identity theorem & isolated zeros. If f, g are holomorphic on a domain U and $f = g$ on a set with a limit point in U, then $f \equiv g$ on U. Consequently, a nonzero holomorphic function on U has zeros that do not accumulate in U.

Path independence and primitives (for Problem 10). If $\omega(z)\,dz$ is holomorphic on a simply connected domain D, then

$$F(z) := \int_{z_0}^{z} \omega(\zeta)\,d\zeta$$

is well defined (independent of path), holomorphic on D, and satisfies $F' = \omega$. In particular, if $0 \notin D$, the 1-form $d\zeta/\zeta$ has the primitive $F(z) = \int_{z_0}^{z} \frac{d\zeta}{\zeta}$ on D, so e^F gives a single-valued branch of the logarithm.

Winding number criterion for a logarithm. A holomorphic, nowhere–vanishing function g on a domain D admits a holomorphic branch of $\operatorname{Log} g$ on D iff $\displaystyle\int_\gamma \frac{g'}{g}\,dz = 0$ for every closed curve $\gamma \subset D$; equivalently, $g(\gamma)$ has winding number 0 about 0 for all such γ. When D is simply connected and g never vanishes, a branch of $\operatorname{Log} g$ always exists.

Power series, radius of convergence, and natural boundaries (for Problem 11). A power series $\sum_{n\geq 0} a_n z^n$ has a radius of convergence $R \in [0, \infty]$ determined by the Cauchy–Hadamard formula. It defines an analytic function on $|z| < R$ and cannot be analytically continued across points where the function has a singularity. A Jordan curve Γ is a *natural boundary* for f if no analytic continuation of f exists across *any* point of Γ.

Hadamard gap theorem (lacunary series). If $f(z) = \sum_{k\geq 1} a_k z^{n_k}$ with strictly increasing exponents (n_k) satisfying $\displaystyle\liminf_{k\to\infty} \frac{n_{k+1}}{n_k} > 1$ ("Hadamard gaps"), then the circle of convergence $|z| = R$ is a *natural boundary* for f (unless f is a polynomial). In particular, for $n_k = 2^k$ (ratio $= 2$) the unit circle is a natural boundary when $R = 1$.

Dense boundary singularities via functional identities (alternate route for Problem 11). If a function f on $|z| < 1$ satisfies identities of the form

$$(1 - z^N)\, f(z) = P_N(z) \qquad (N \in \mathbb{N}),$$

with P_N polynomial and $N \to \infty$, then every N-th root of unity is a singularity (unless canceled), and the union of these roots for $N \to \infty$ is dense on $|z| = 1$. Hence $|z| = 1$ is a natural boundary (no analytic continuation across any boundary point).

Problem. Let f be holomorphic on $D(a, R)$. If w satisfies $|w - a| < r < R$, show

$$f^{(n)}(w) = \frac{n!}{2\pi i} \int_{|z-a|=r} \frac{f(z)}{(z - w)^{n+1}}\,dz.$$

Background. *Residue theorem.* If g is meromorphic on and inside a positively oriented simple closed curve γ (no poles on γ),

$$\int_\gamma g(z)\,dz = 2\pi i \sum \operatorname{Res}(g; a_k).$$

Higher–order pole. If g has a pole of order m at w,

$$\operatorname{Res}(g; w) = \frac{1}{(m-1)!} \lim_{z\to w} \frac{d^{m-1}}{dz^{m-1}} \Big((z-w)^m g(z) \Big).$$

Taylor series. $f(z) = \sum_{k\geq 0} \dfrac{f^{(k)}(w)}{k!}(z-w)^k$ near w.

Solution (via residues). Let $g(z) = \dfrac{f(z)}{(z-w)^{n+1}}$. Then g is meromorphic on $|z-a| \leq r$ with a single pole of order $n+1$ at $z = w$. Hence

$$\operatorname{Res}(g; w) = \frac{1}{n!} f^{(n)}(w).$$

Applying the residue theorem to the circle $|z-a| = r$ gives

$$\int_{|z-a|=r} \frac{f(z)}{(z-w)^{n+1}}\,dz = 2\pi i \operatorname{Res}(g; w) = 2\pi i \frac{f^{(n)}(w)}{n!},$$

which rearranges to the stated formula. For $n = 0$ this is the usual Cauchy integral formula.

Examples.

- $f(z) = e^z$: $\dfrac{n!}{2\pi i} \displaystyle\int_{|z-a|=r} \frac{e^z}{(z-w)^{n+1}}\,dz = e^w.$
- $f(z) = \sum_{k=0}^m a_k z^k$: the integral vanishes for $n > m$, agreeing with $f^{(n)} \equiv 0$.
- The integral is independent of r as long as $|w-a| < r < R$ (no singularities crossed when deforming the contour).

$\square$

Pole of order 3: coefficients c_{-3}, c_{-2}, c_{-1} and the residue. Suppose f has a pole of order 3 at a. Define

$$h(z) := (z-a)^3 f(z) = \sum_{n=0}^\infty d_n (z-a)^n, \qquad d_n = \frac{h^{(n)}(a)}{n!}.$$

Then

$$f(z) = \frac{d_0}{(z-a)^3} + \frac{d_1}{(z-a)^2} + \frac{d_2}{(z-a)} + \cdots,$$

so

$$\boxed{c_{-3} = d_0, \qquad c_{-2} = d_1, \qquad c_{-1} = d_2 = \operatorname{Res}(f; a)}.$$

Example. For $f(z) = \dfrac{e^z}{(z-a)^3}$ we have $h(z) = e^z$, hence

$$c_{-3} = e^a, \qquad c_{-2} = e^a, \qquad c_{-1} = \frac{e^a}{2}, \qquad \operatorname{Res}(f; a) = \frac{e^a}{2}.$$

Equivalently, $\operatorname{Res}(f; a) = \dfrac{1}{2!} \dfrac{d^2}{dz^2} \big((z-a)^3 f(z) \big)\Big|_{z=a}$.

Three distinct poles inside a contour: sum of residues. Let γ be a positively oriented simple closed curve and assume f is meromorphic on and inside γ, with distinct poles a, b, c inside and none on γ. Then the residue theorem gives

$$\boxed{\int_\gamma f(z)\,dz \;=\; 2\pi i\big(\operatorname{Res}(f;a) + \operatorname{Res}(f;b) + \operatorname{Res}(f;c)\big).}$$

Simple-pole residues. If $f = \dfrac{p}{q}$ with p, q holomorphic and $q'(a) \neq 0$,

$$\operatorname{Res}(f;a) = \frac{p(a)}{q'(a)}.$$

For example, for $f(z) = \dfrac{1}{(z-a)(z-b)(z-c)}$,

$$\operatorname{Res}(f;a) = \frac{1}{(a-b)(a-c)}, \quad \operatorname{Res}(f;b) = \frac{1}{(b-a)(b-c)}, \quad \operatorname{Res}(f;c) = \frac{1}{(c-a)(c-b)},$$

and hence

$$\int_\gamma f(z)\,dz = 2\pi i\left(\frac{1}{(a-b)(a-c)} + \frac{1}{(b-a)(b-c)} + \frac{1}{(c-a)(c-b)}\right).$$